

Yellow River Marsh Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

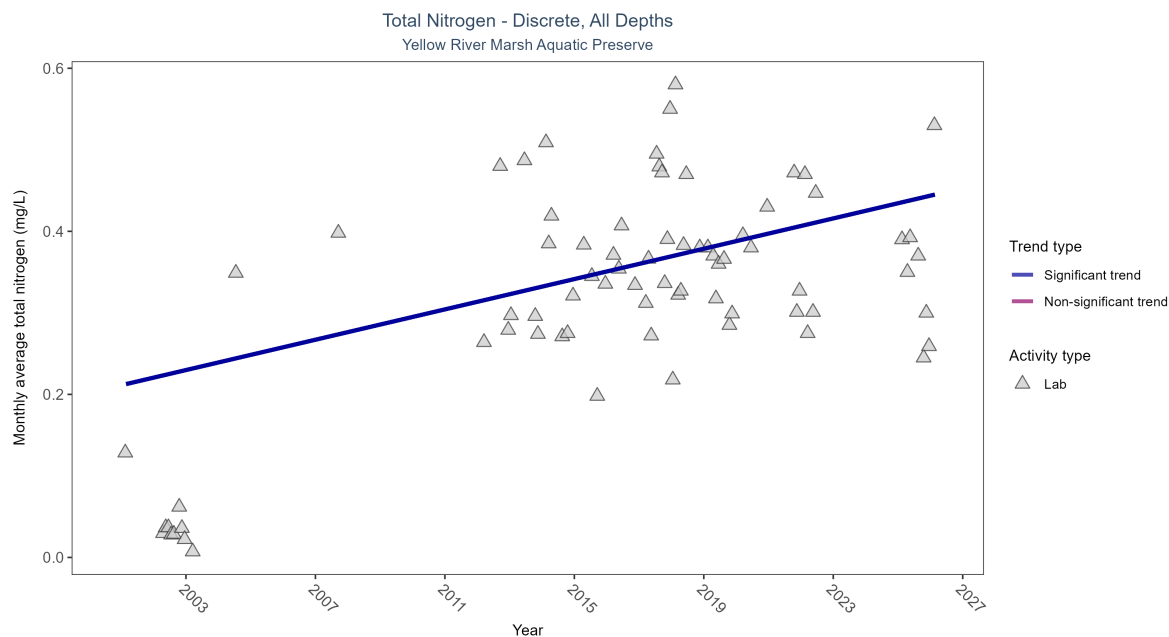


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen increased by 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	117	19	2001 - 2026	0.327	0.19492	0.21123	0.0093	0.036

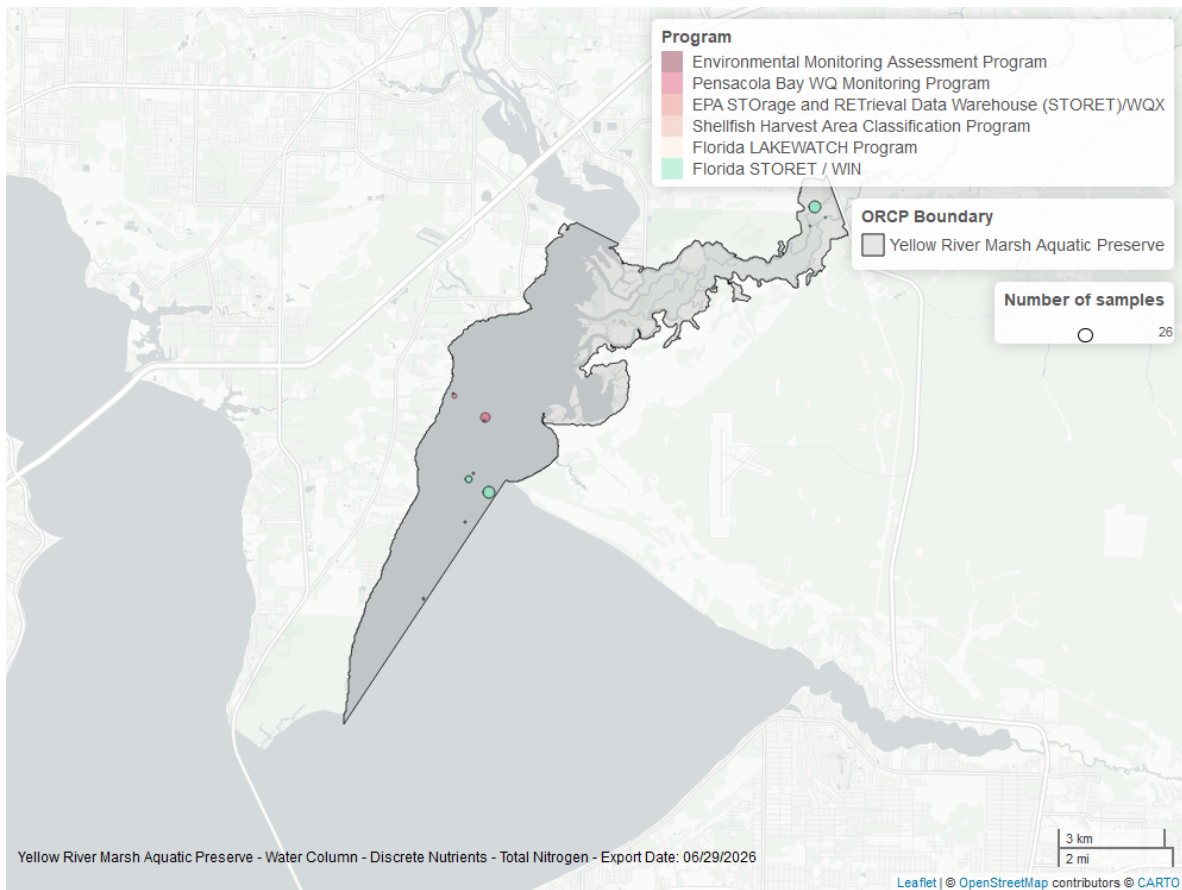


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

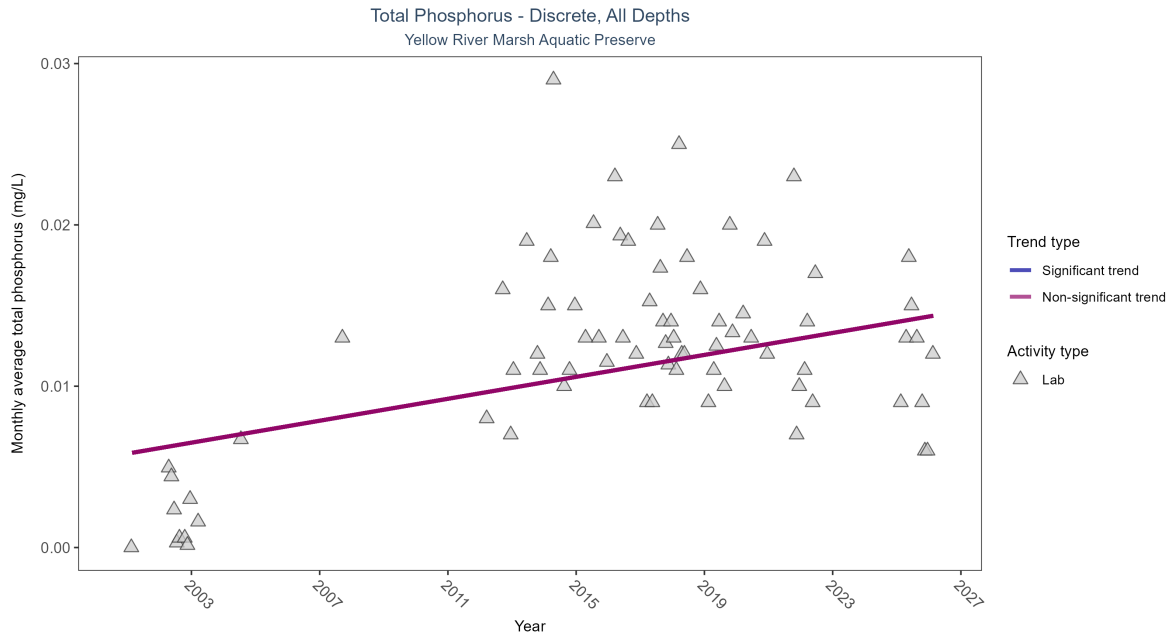


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Total phosphorus showed no detectable trend between 2001 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	113	18	2001 - 2026	0.012	0.21317	0.00582	0.00034	0.0689

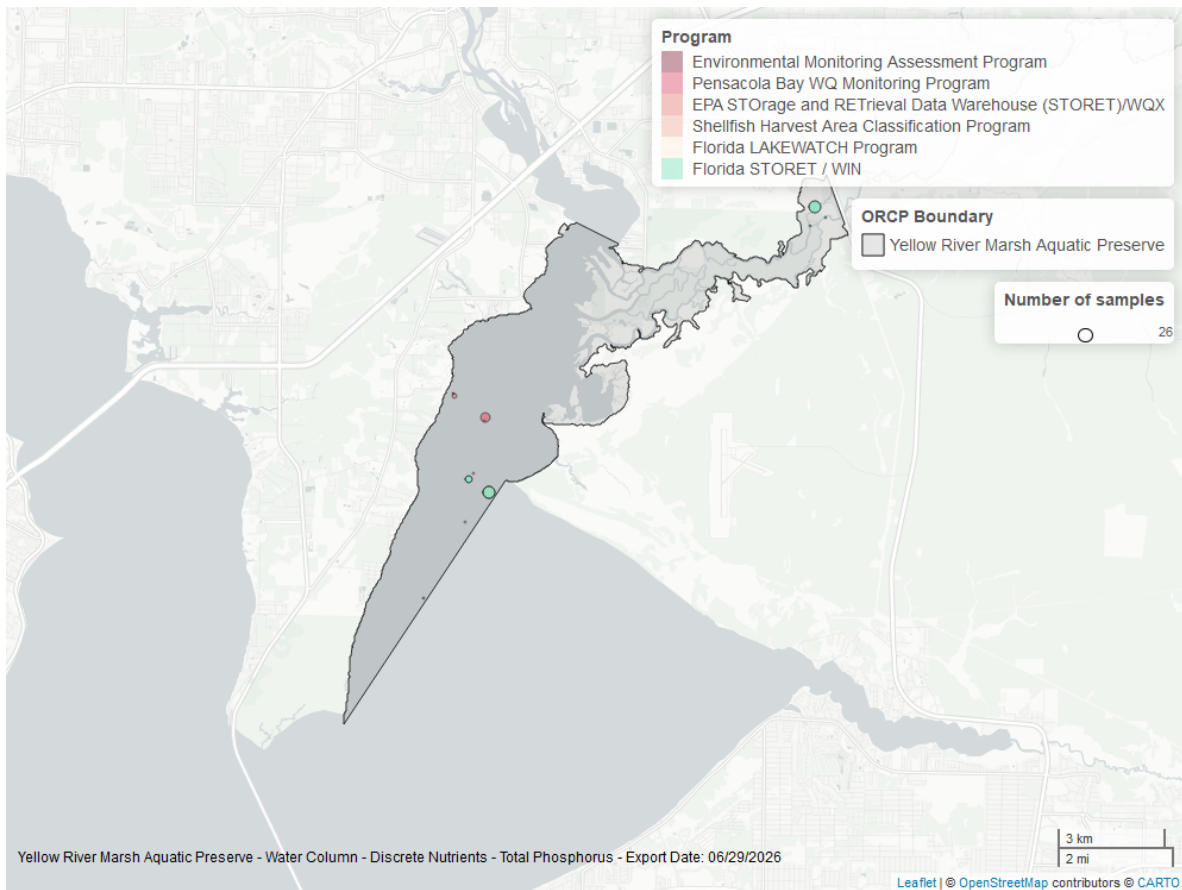


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

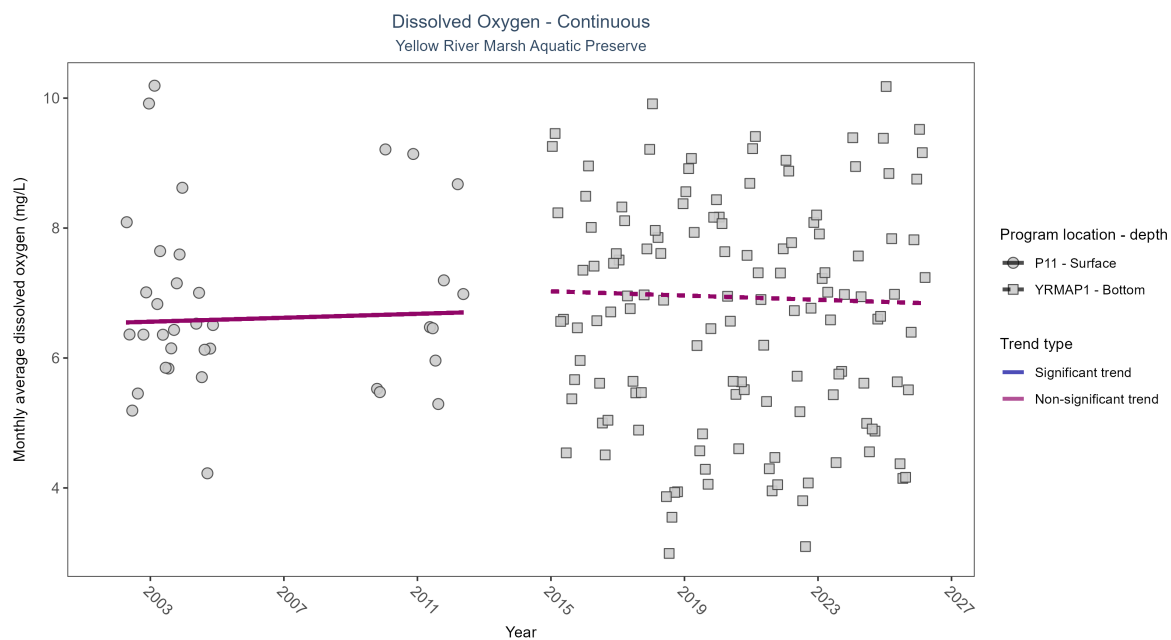


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: No detectable change in monthly average dissolved oxygen was observed at two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	No detectable trend	335729	12	2015 - 2026	6.90	-0.08	7.03	-0.02	0.23
P11	No detectable trend	131	7	2002 - 2012	6.37	0.04	6.54	0.02	0.49

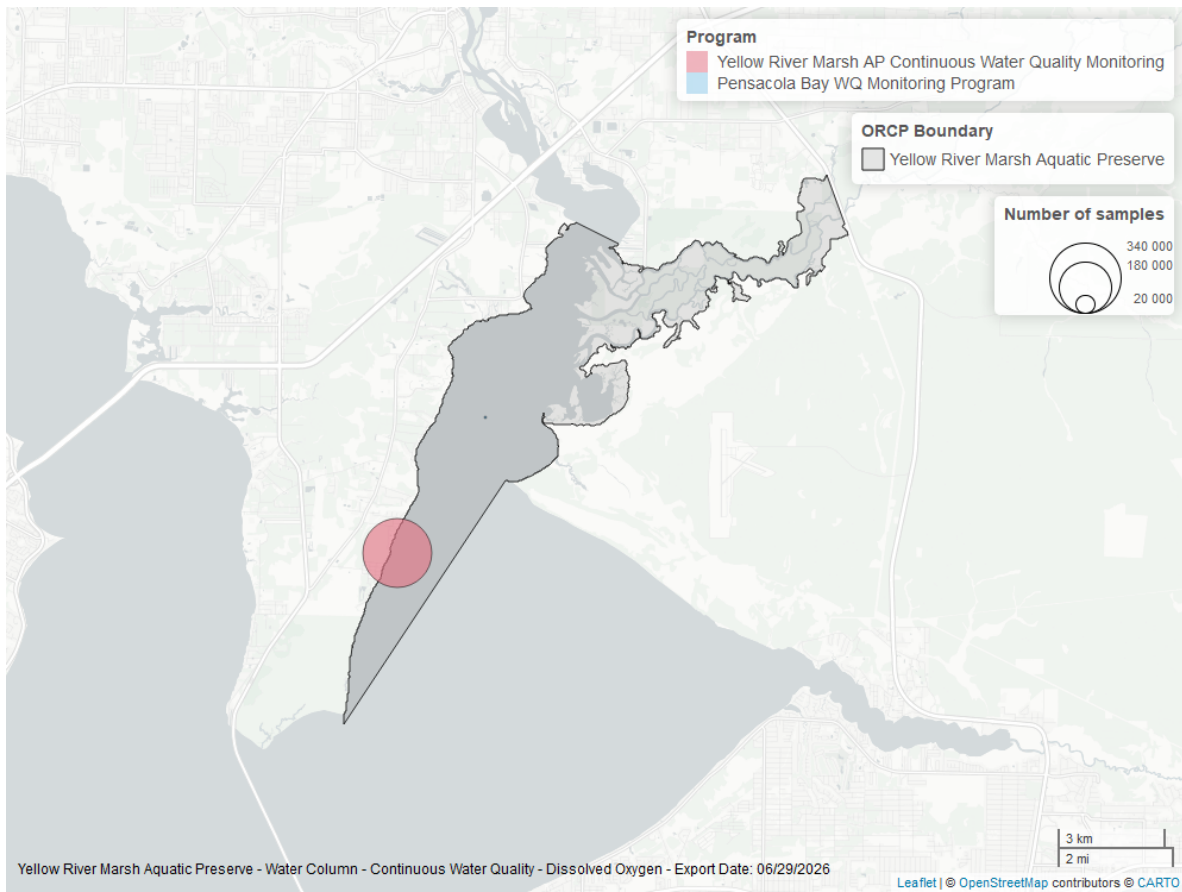


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

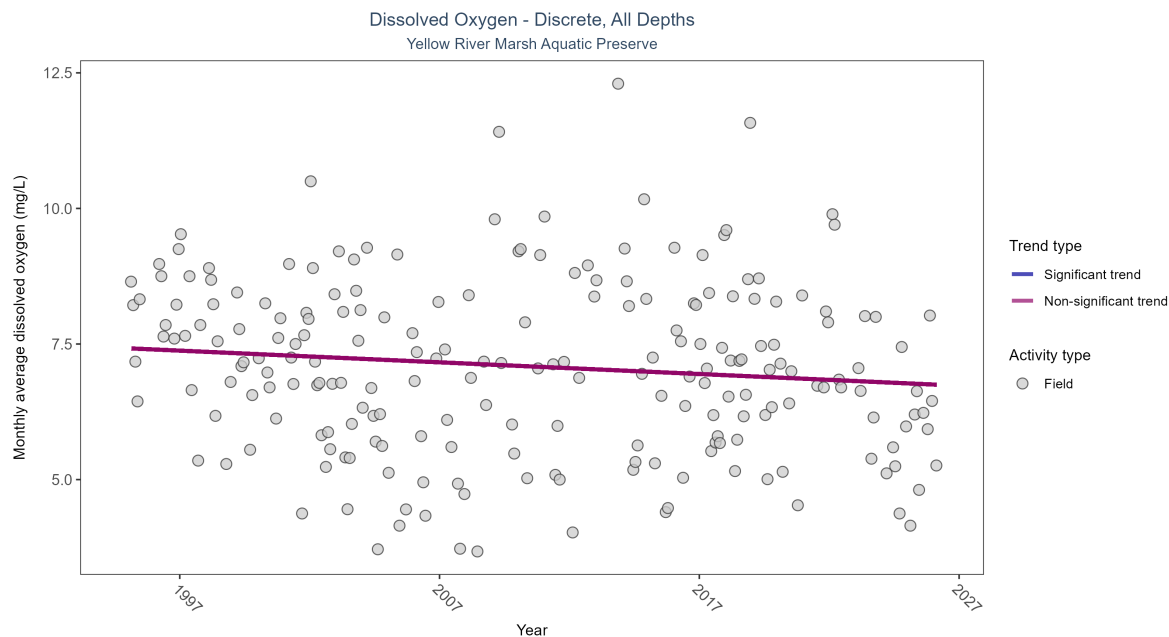


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Dissolved oxygen showed no detectable trend between 1995 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	1090	32	1995 - 2026	7.2	-0.06914	7.42086	-0.02155	0.0536

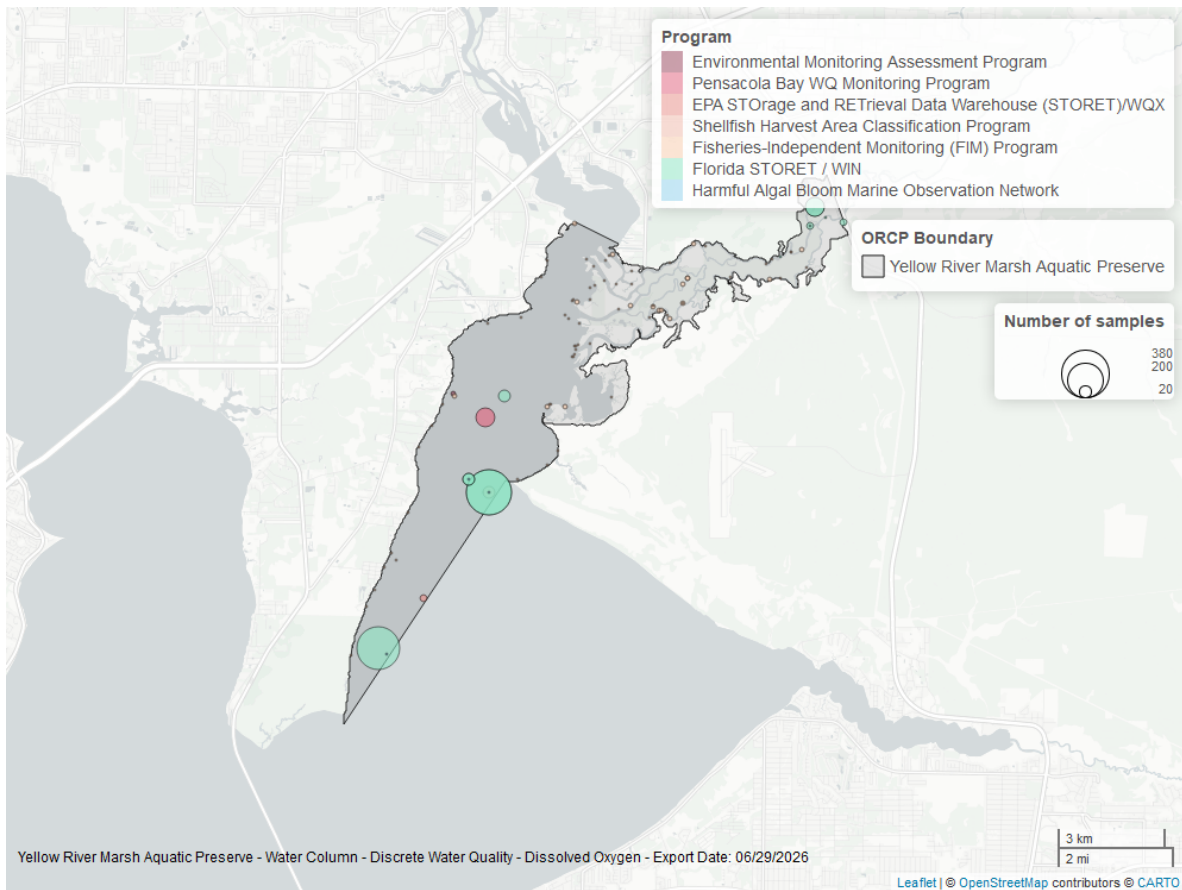


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

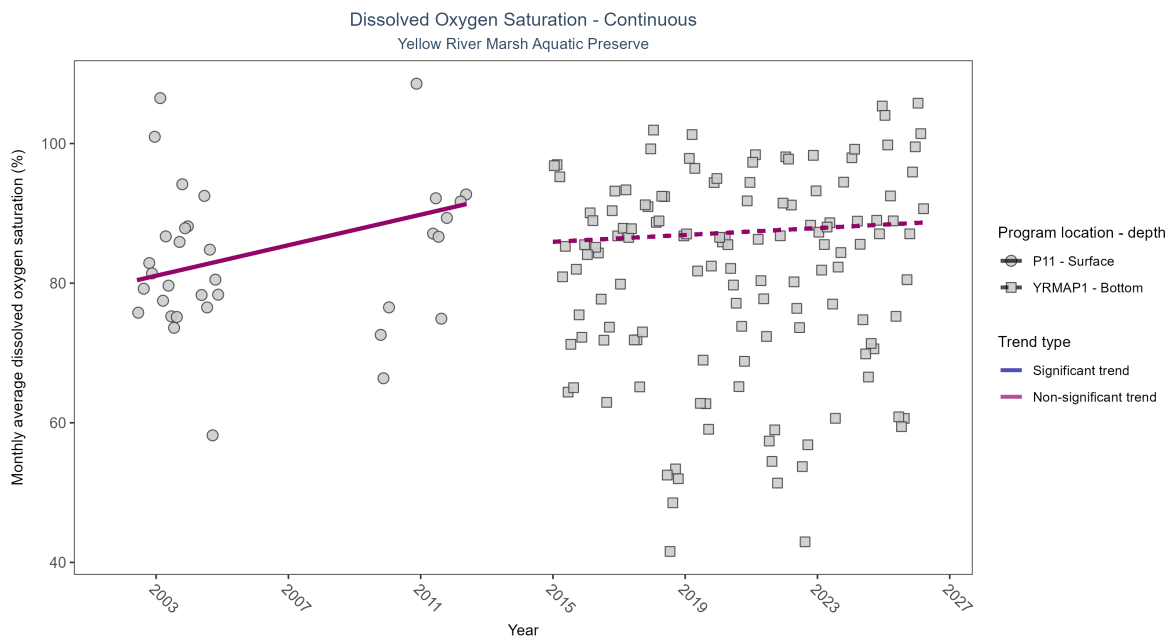


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: No detectable change in monthly average dissolved oxygen saturation was observed at two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	No detectable trend	342018	12	2015 - 2026	87.30	0.07	85.92	0.25	0.32
P11	No detectable trend	126	7	2002 - 2012	79.93	0.09	79.97	1.09	0.71

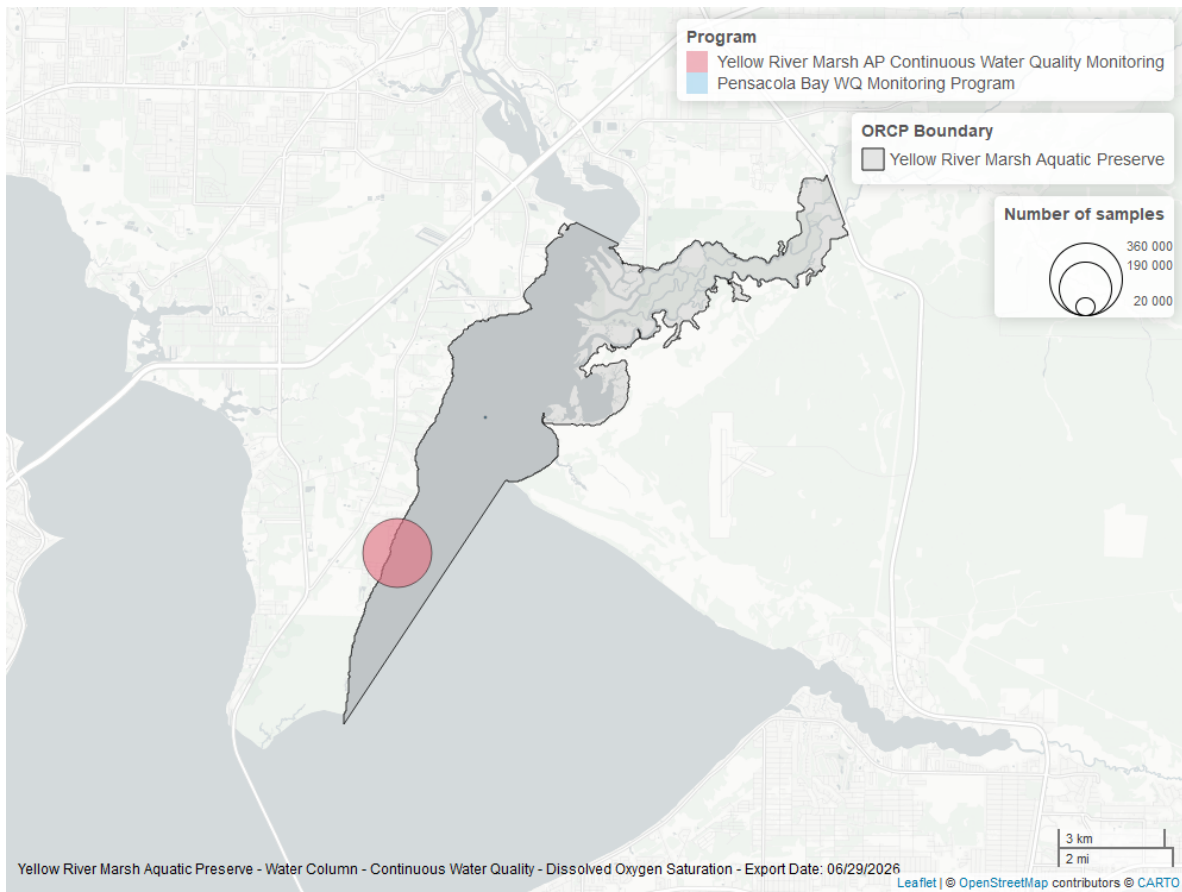


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

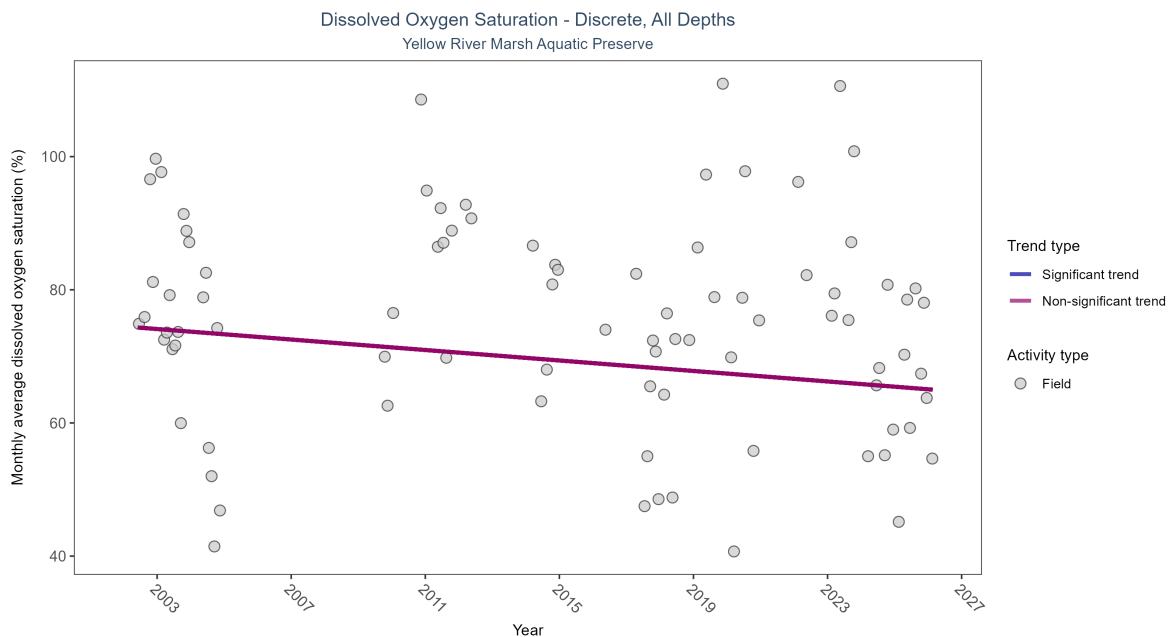


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Dissolved oxygen saturation showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	195	18	2002 - 2026	75.2	-0.11297	74.49593	-0.39353	0.1686

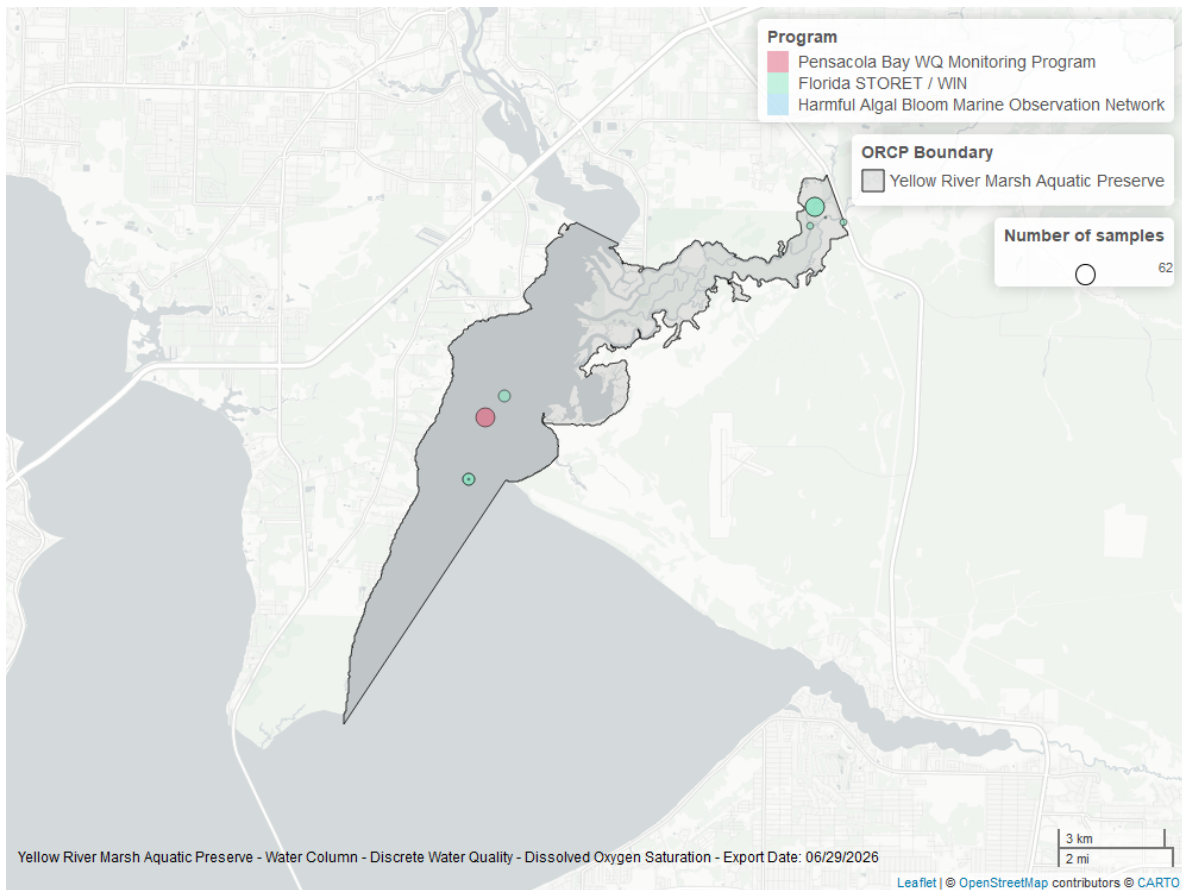


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

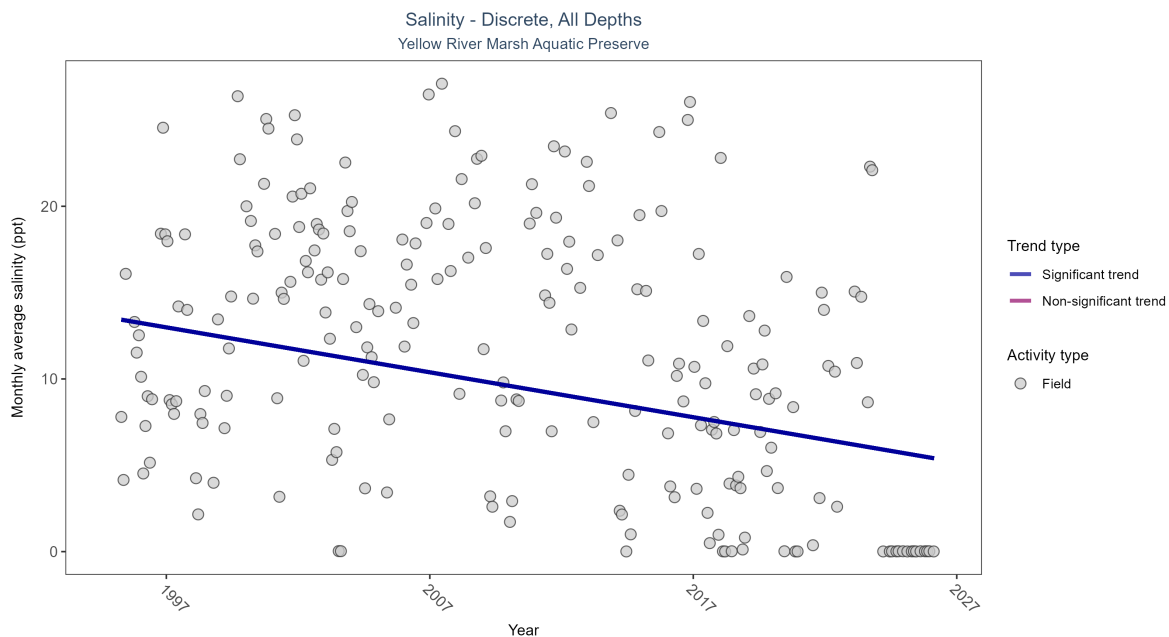


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Monthly average salinity decreased by 0.26 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	1212	32	1995 - 2026	10.9	-0.19556	13.50744	-0.26017	0

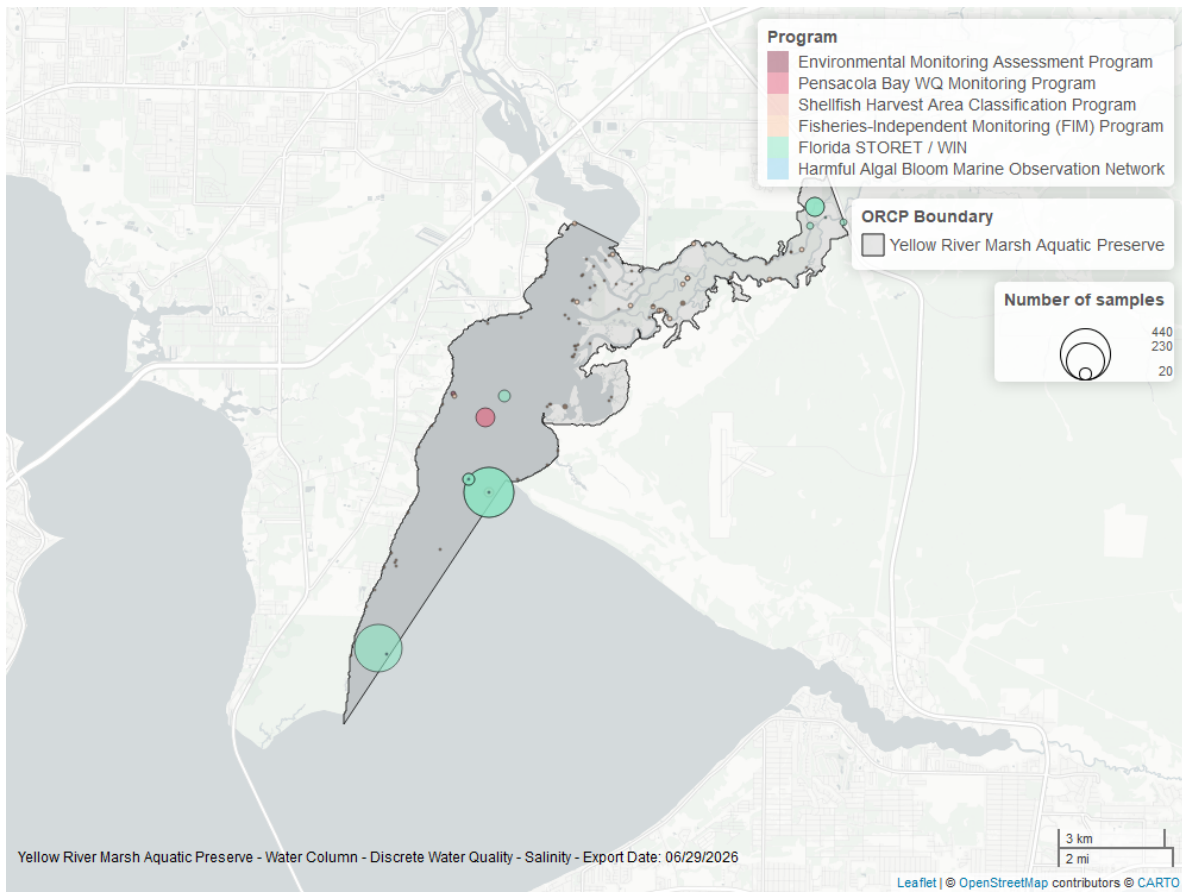


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

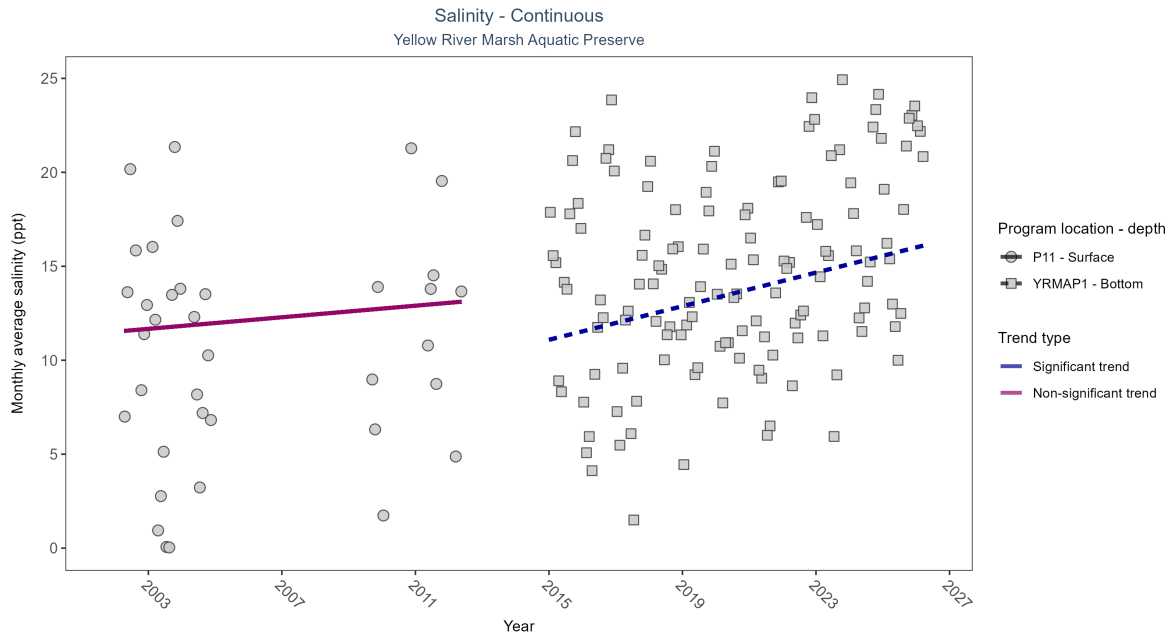


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: At one program location, monthly average salinity increased by 0.45 ppt per year. No detectable change in monthly average salinity was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	Increasing trend	348400	12	2015 - 2026	14.80	0.30	11.09	0.45	0.00
P11	No detectable trend	136	7	2002 - 2012	10.05	0.06	11.52	0.15	0.65

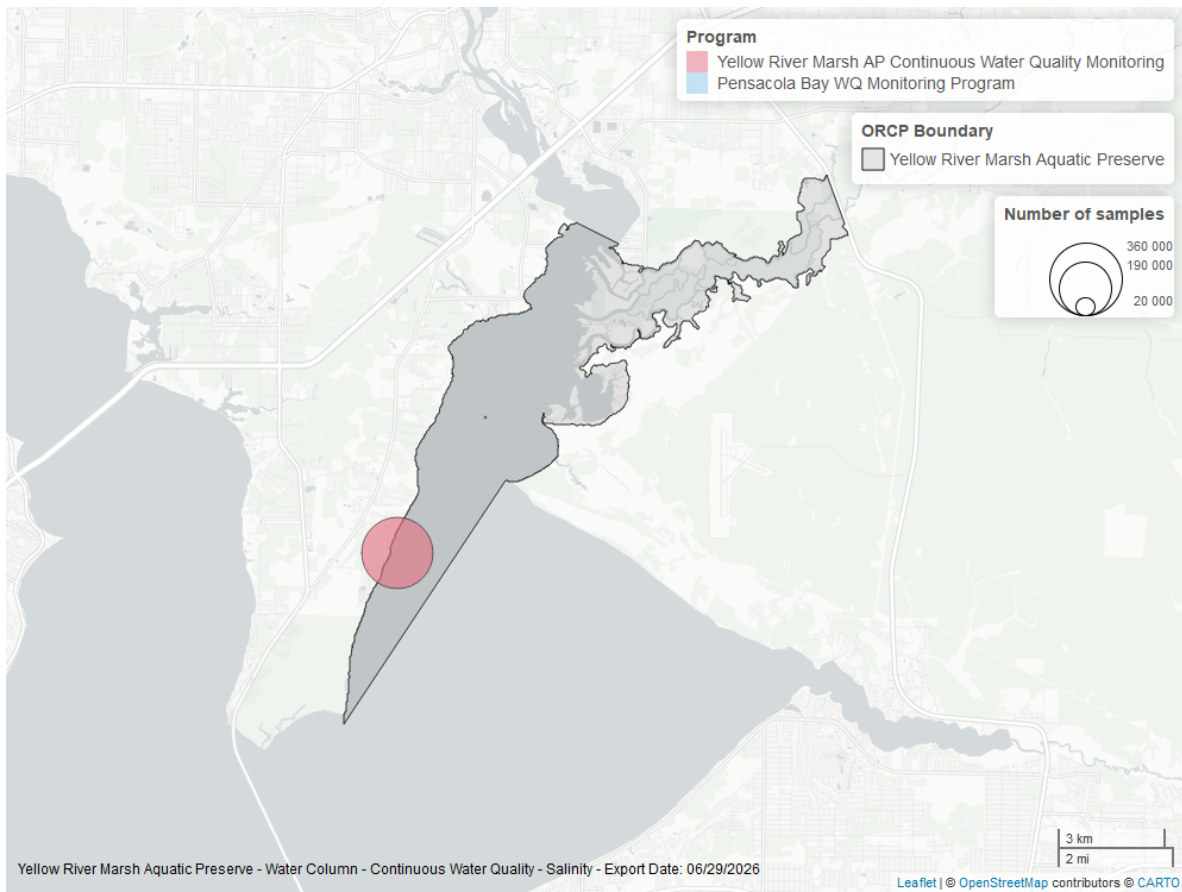


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

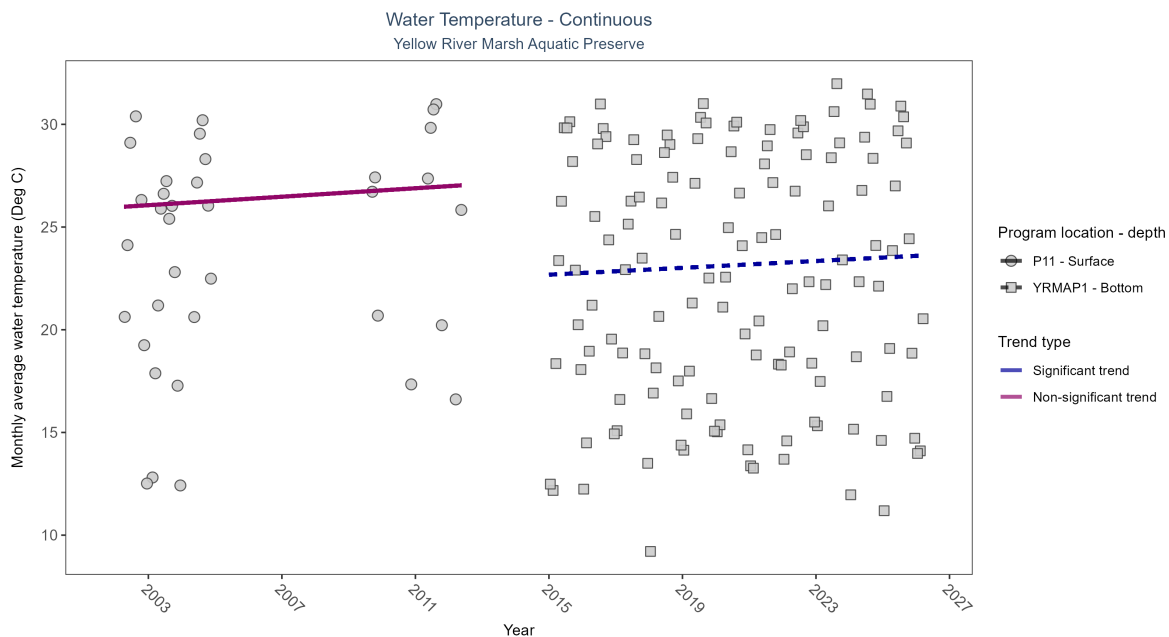


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: At one program location, monthly average water temperature increased by 0.08Deg C per year. No detectable change in monthly average water temperature was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	Increasing trend	379144	12	2015 - 2026	22.90	0.15	22.68	0.08	0.03
P11	No detectable trend	136	7	2002 - 2012	26.22	0.13	25.96	0.10	0.50

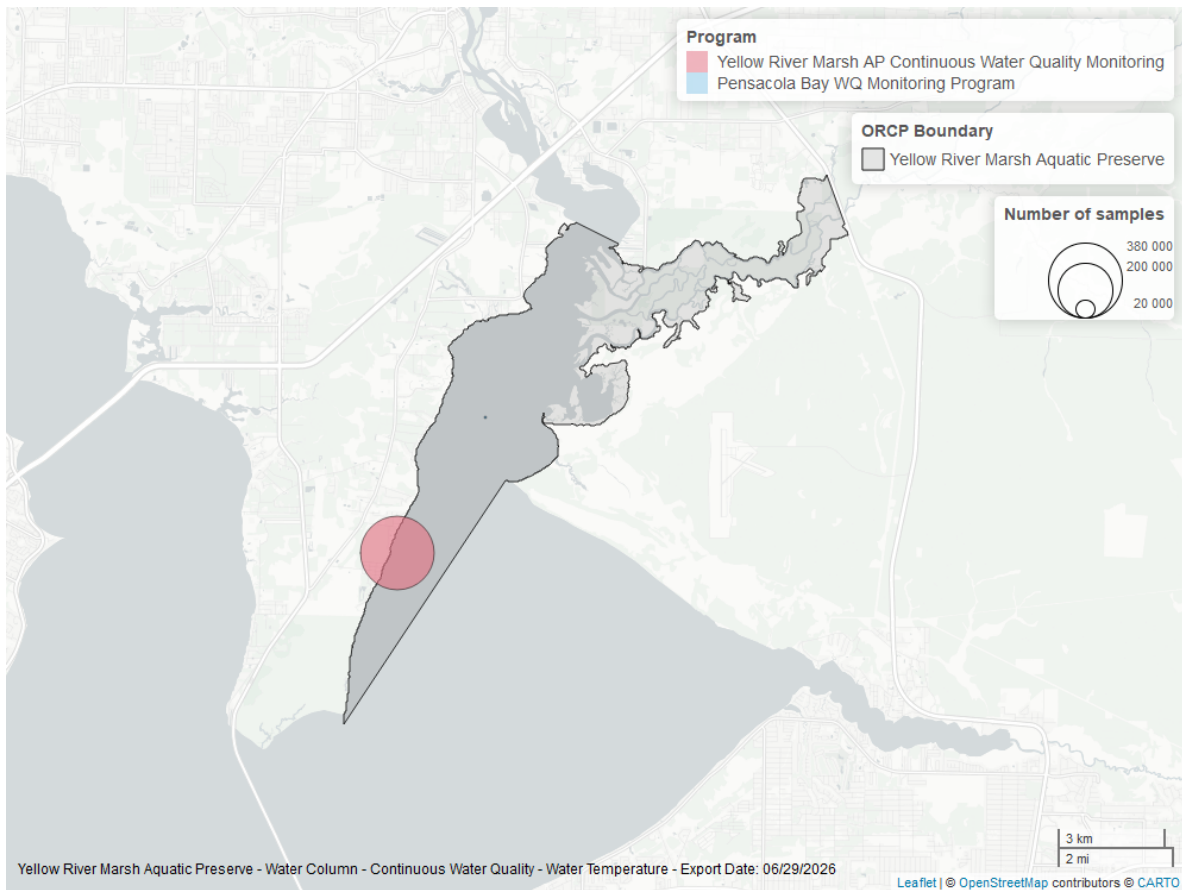


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

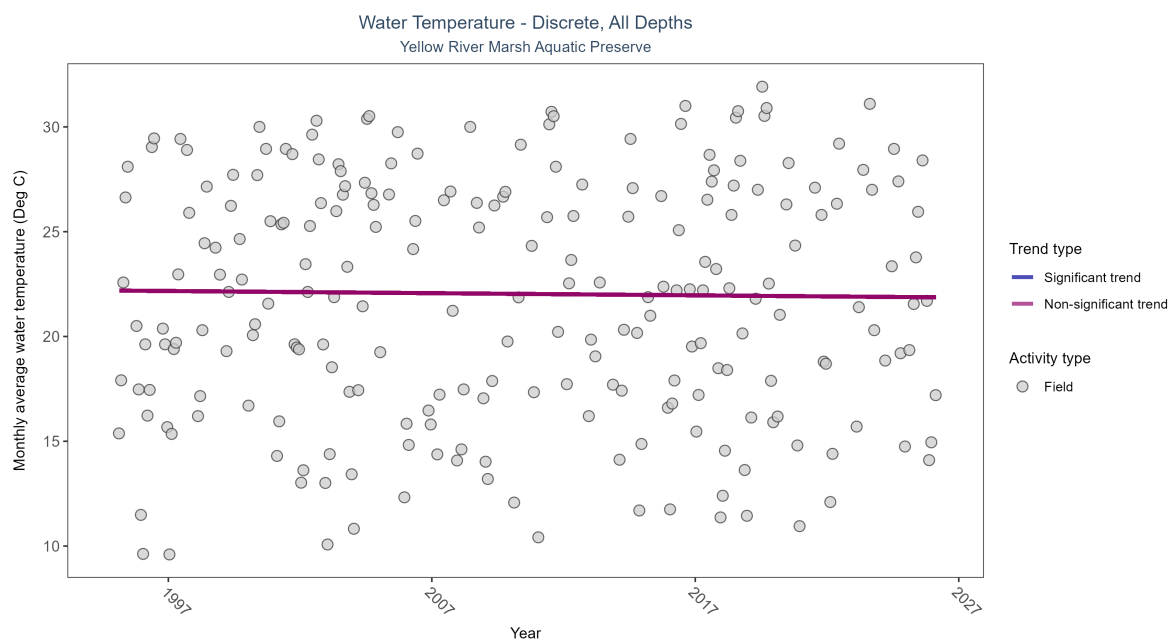


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Water temperature showed no detectable trend between 1995 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	1257	32	1995 - 2026	22	-0.01952	22.19026	-0.01033	0.556

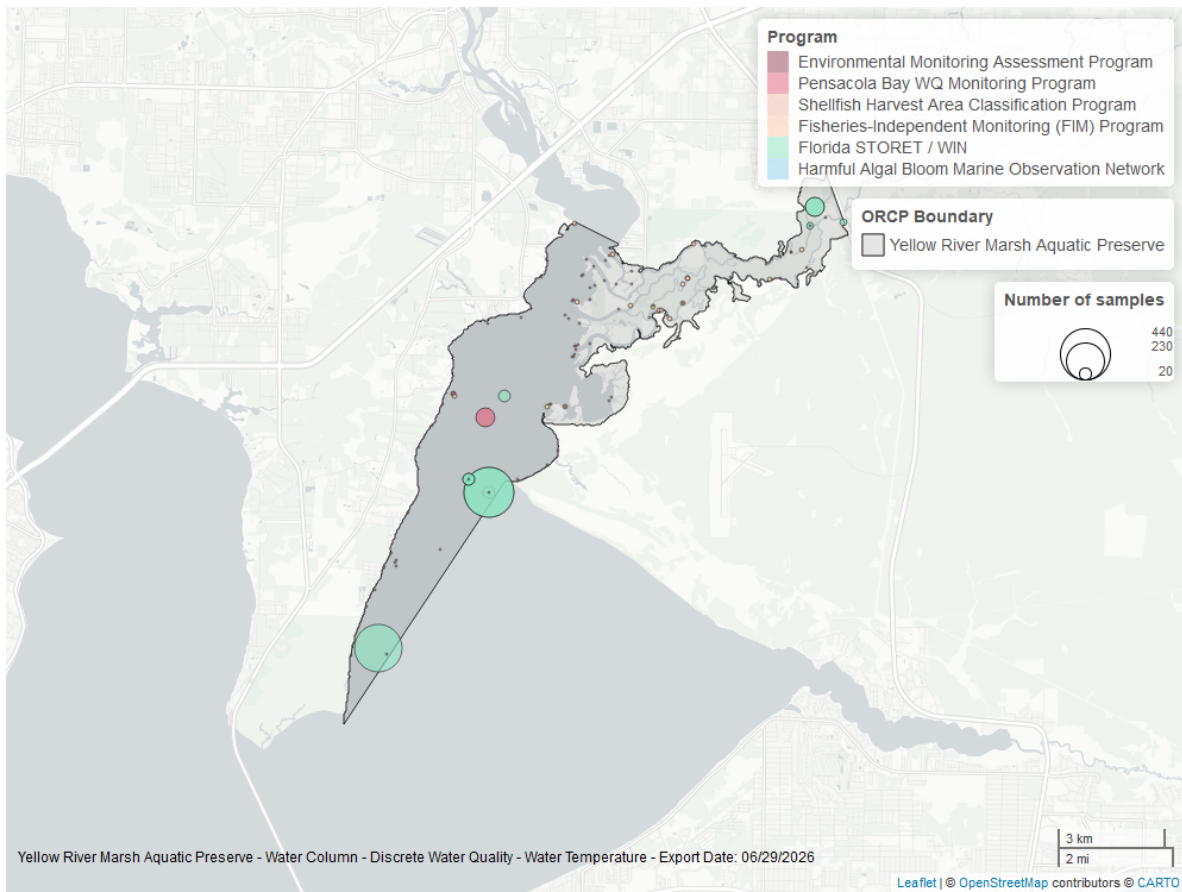


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

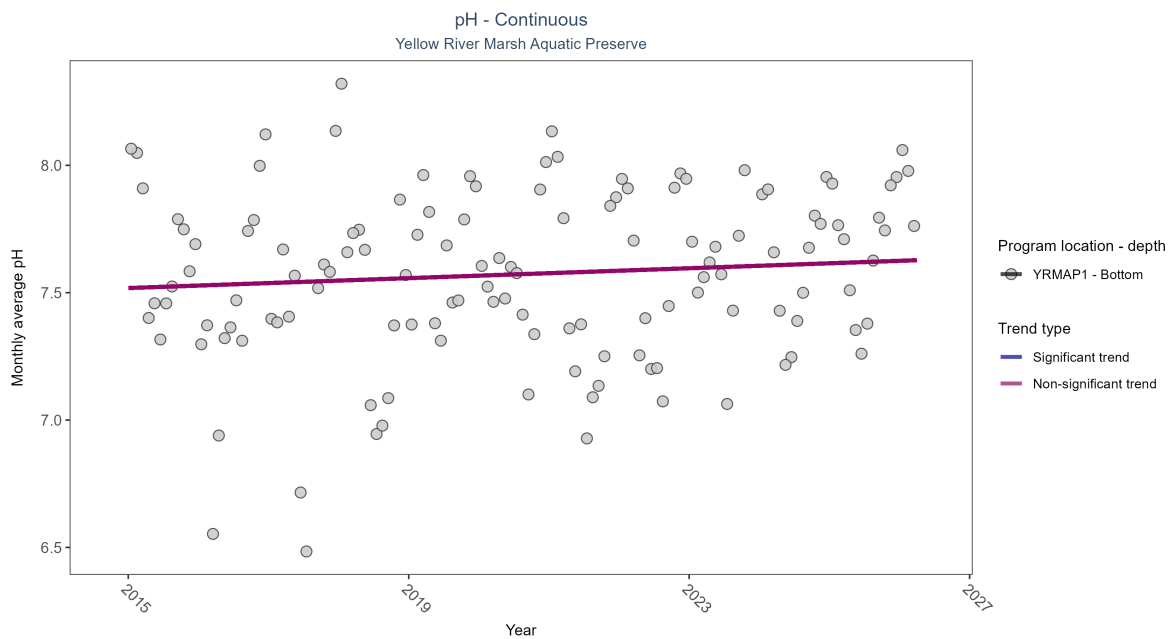


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: No detectable change in monthly average pH was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	No detectable trend	347831	12	2015 - 2026	7.6	0.09	7.52	0.01	0.18

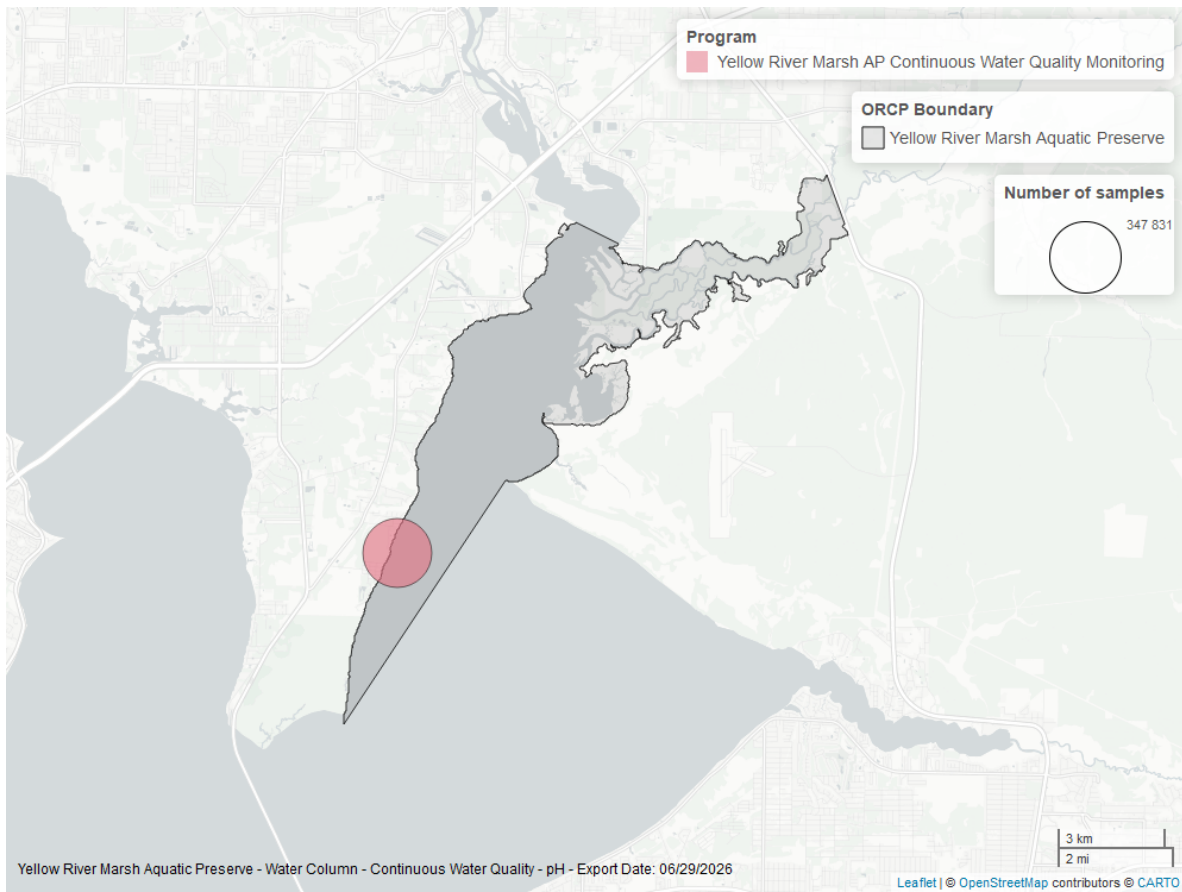


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

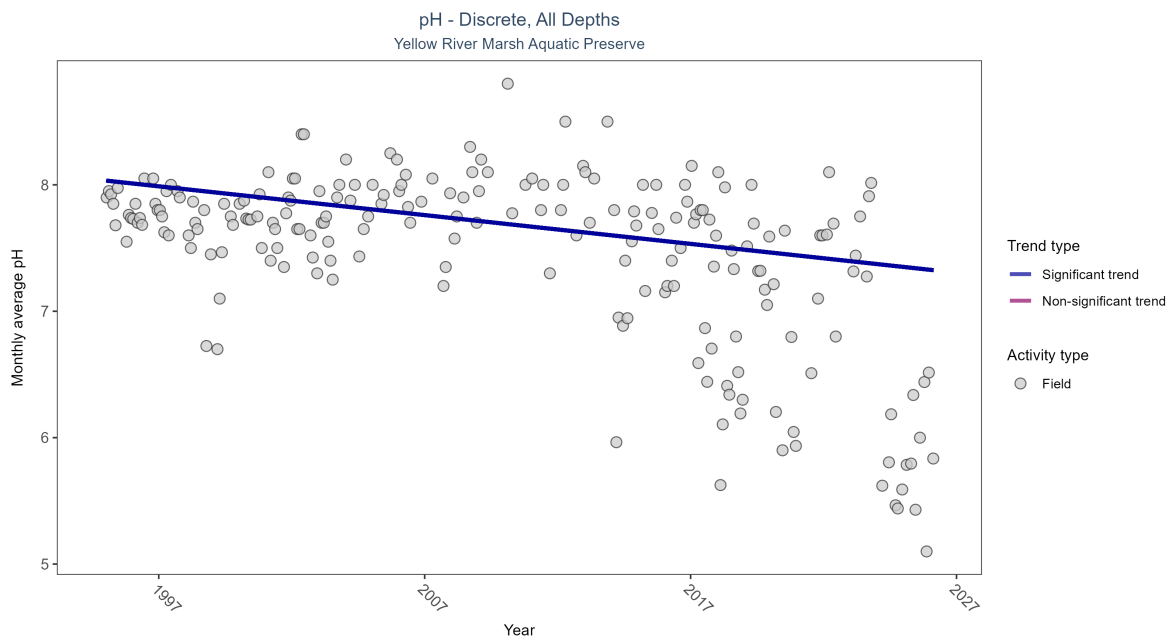


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Monthly average pH decreased by 0.02 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	763	32	1995 - 2026	7.6	-0.23866	8.03381	-0.02277	0

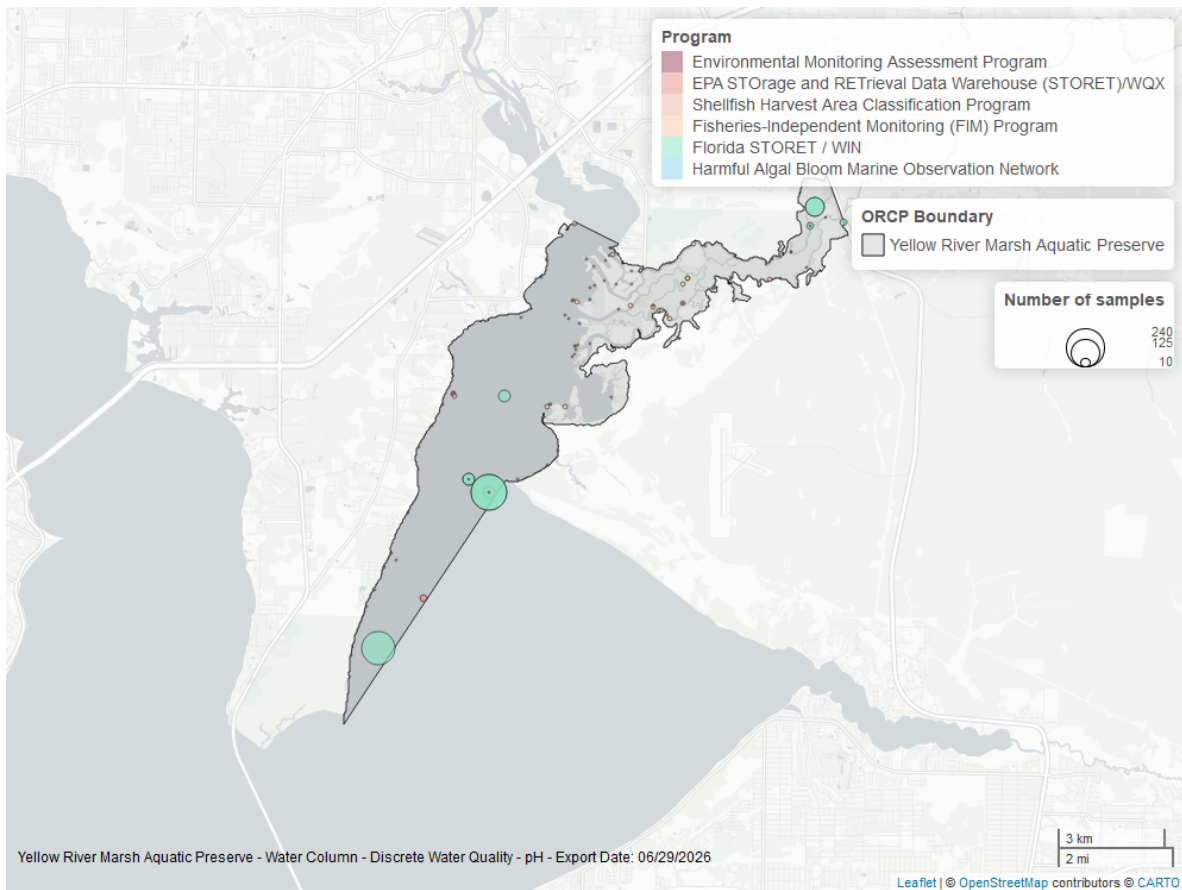


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

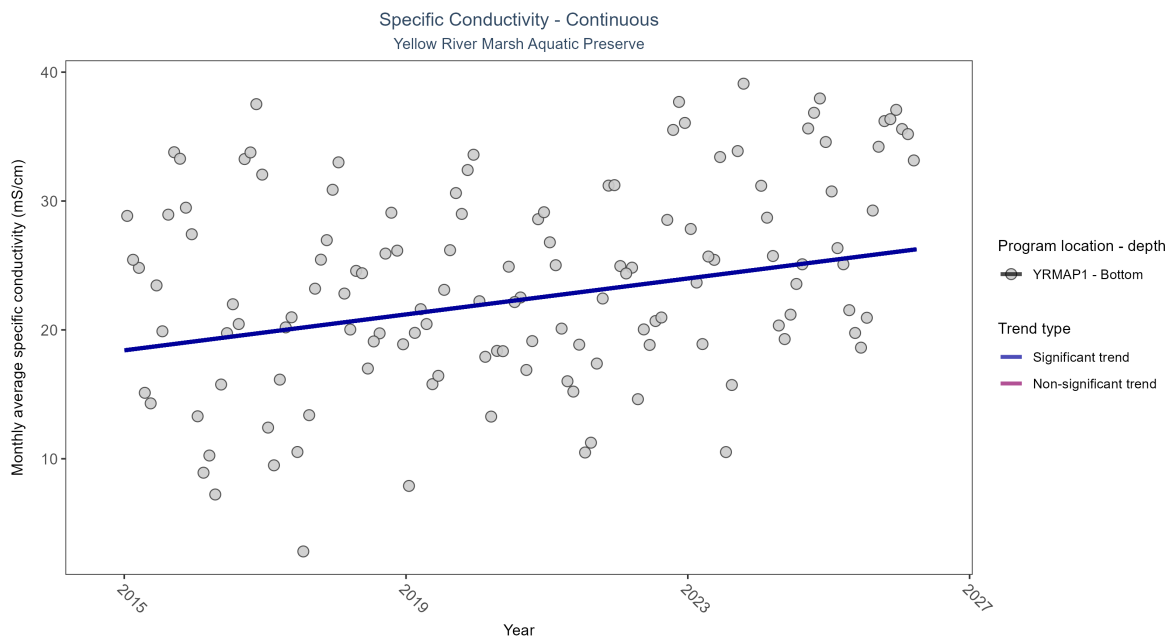


Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: At one program location, monthly average specific conductivity increased by 0.70 mS/cm per year.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	Increasing trend	353373	12	2015 - 2026	24.49	0.32	18.41	0.7	0

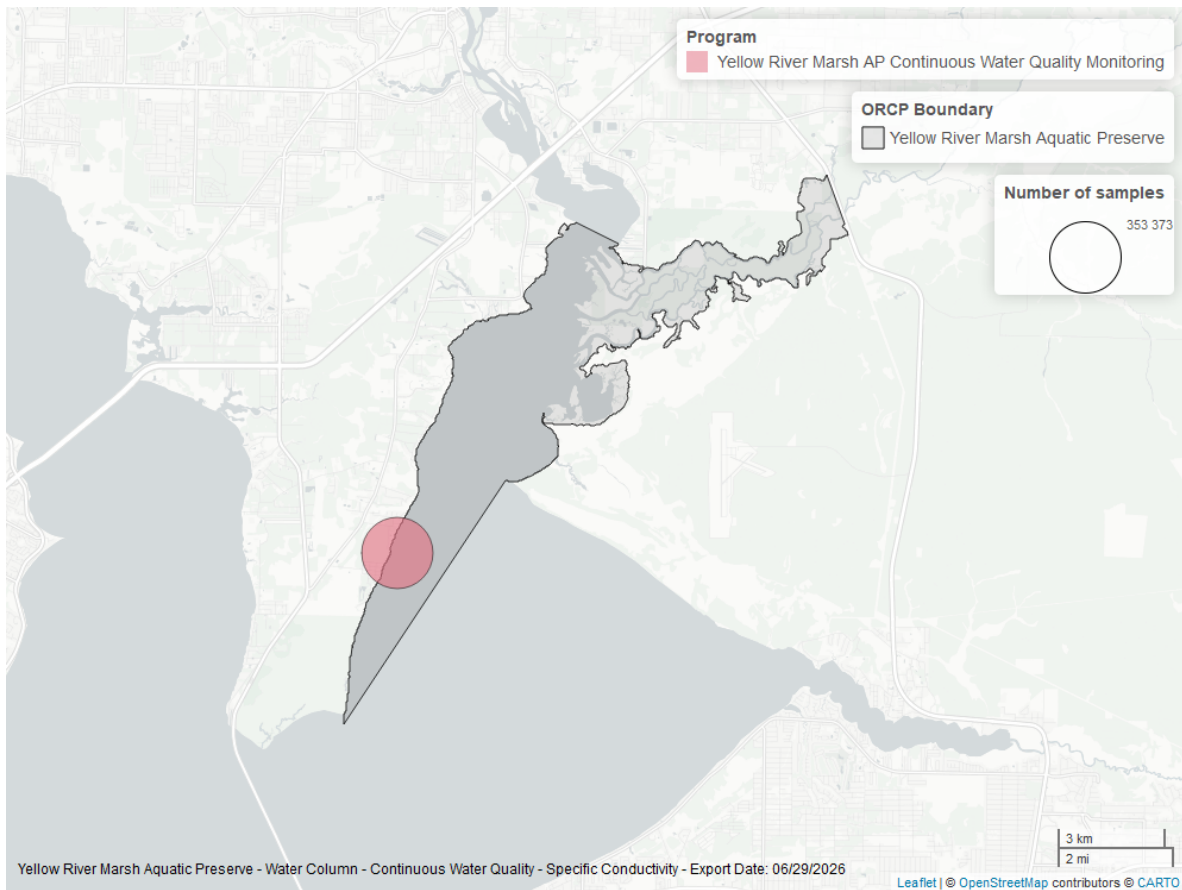


Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

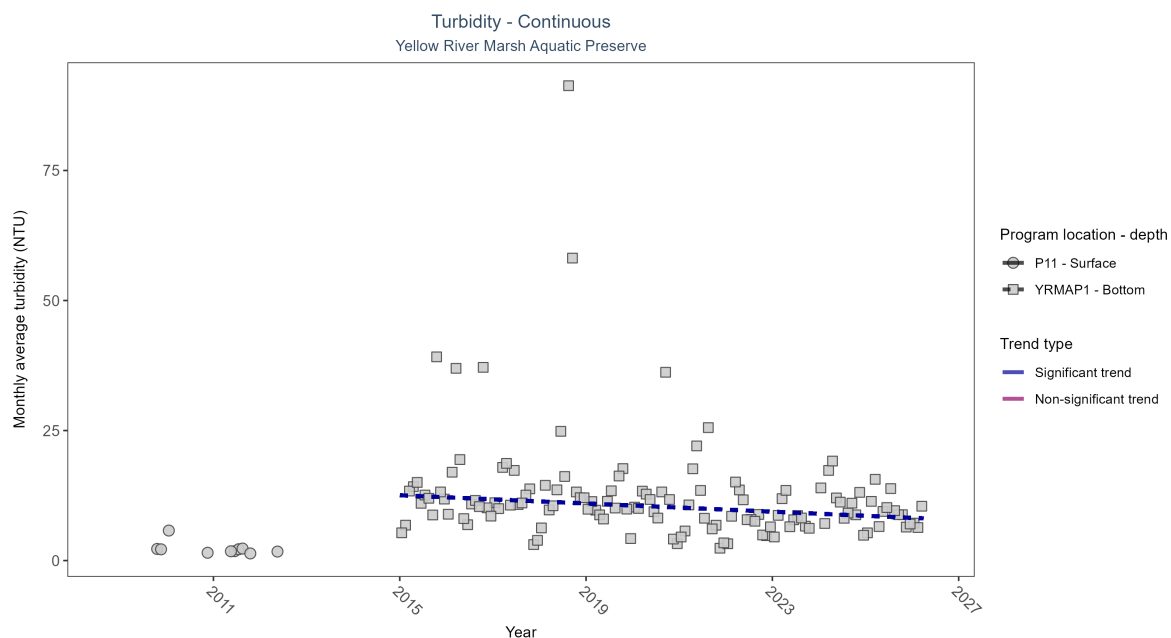


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: At one program location, monthly average turbidity decreased by 0.39 NTU per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	Decreasing trend	361848	12	2015 - 2026	6	-0.22	12.56	-0.39	0
P11	Insufficient data	37	4	2009 - 2012	2	-	-	-	-

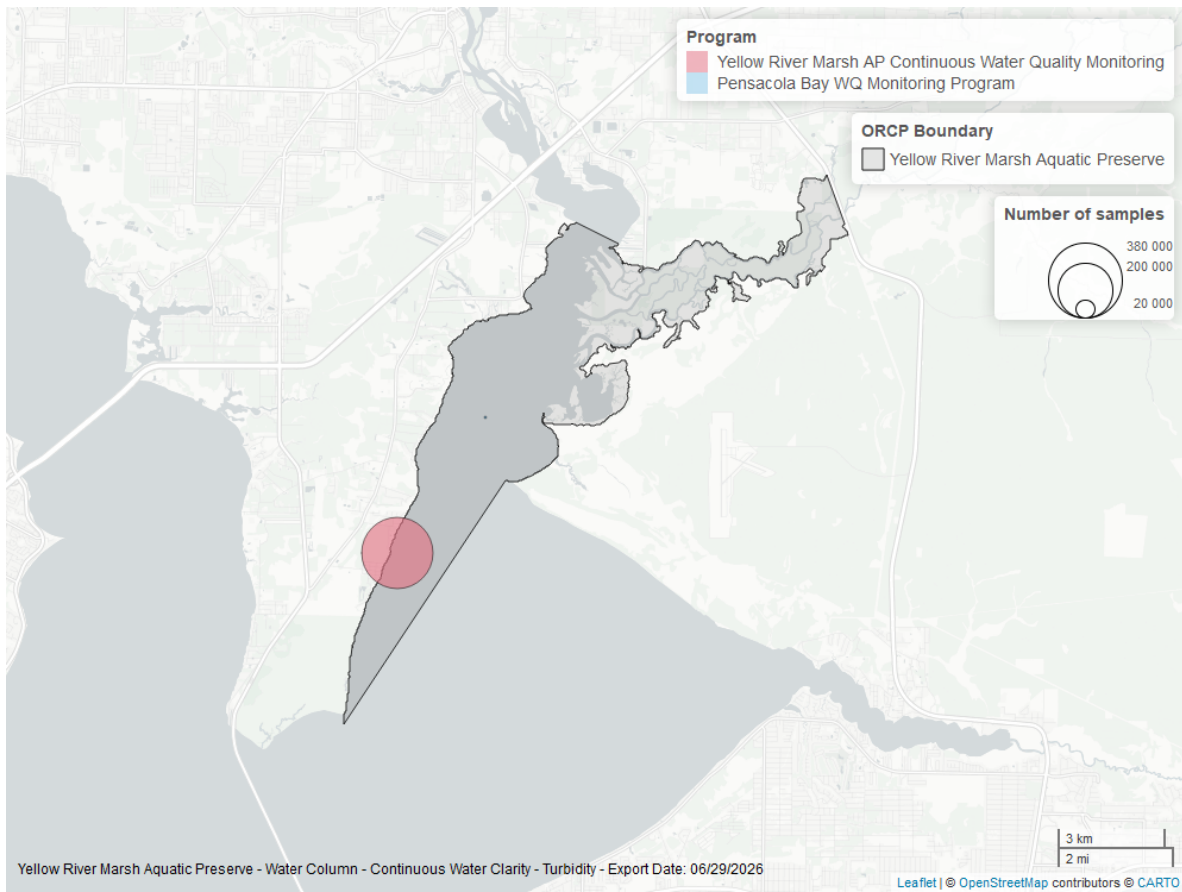


Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

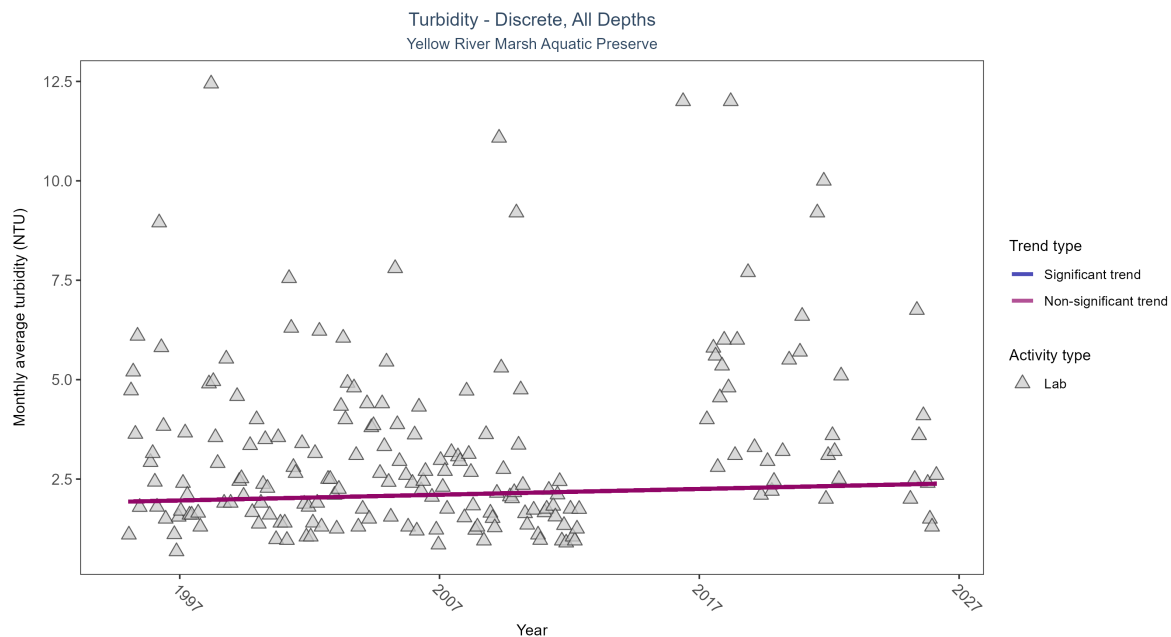


Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Turbidity showed no detectable trend between 1995 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	599	27	1995 - 2026	2.5	0.07659	1.93286	0.01447	0.4083

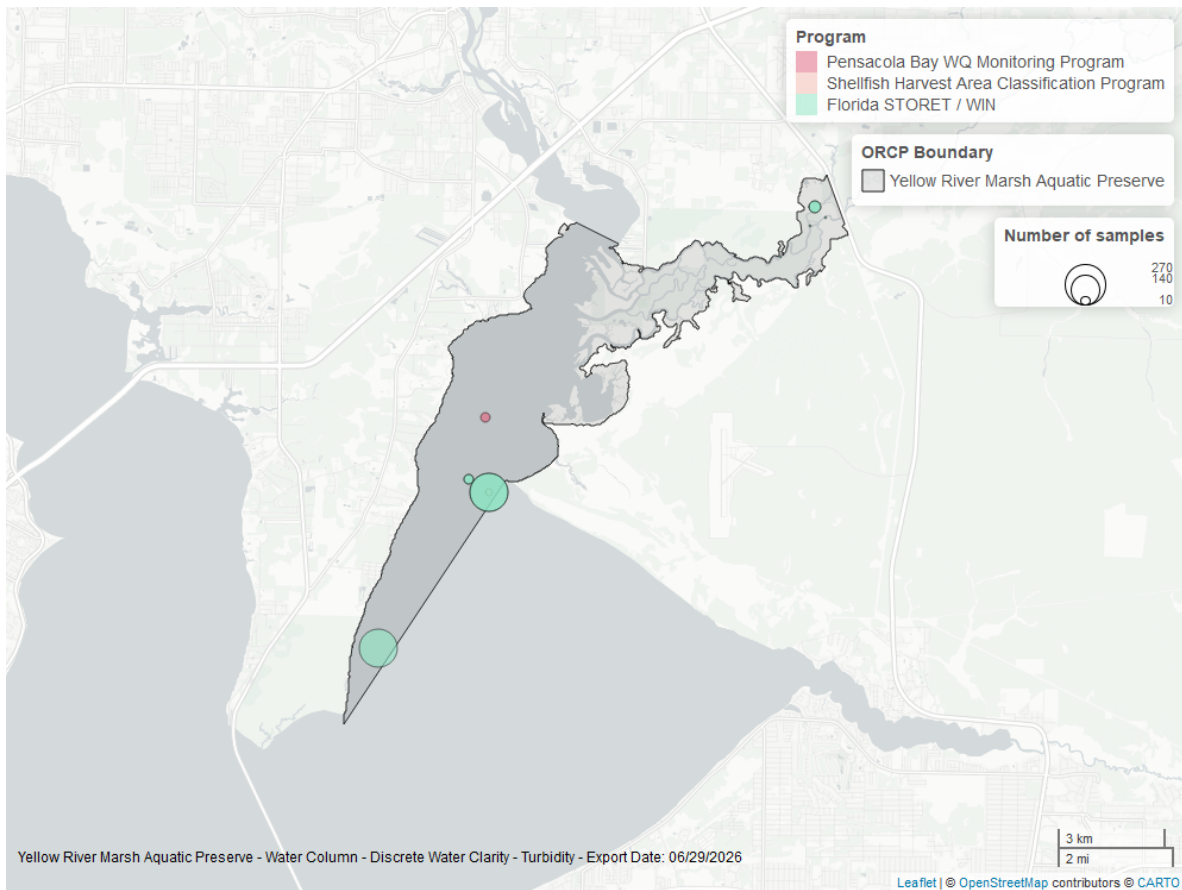


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

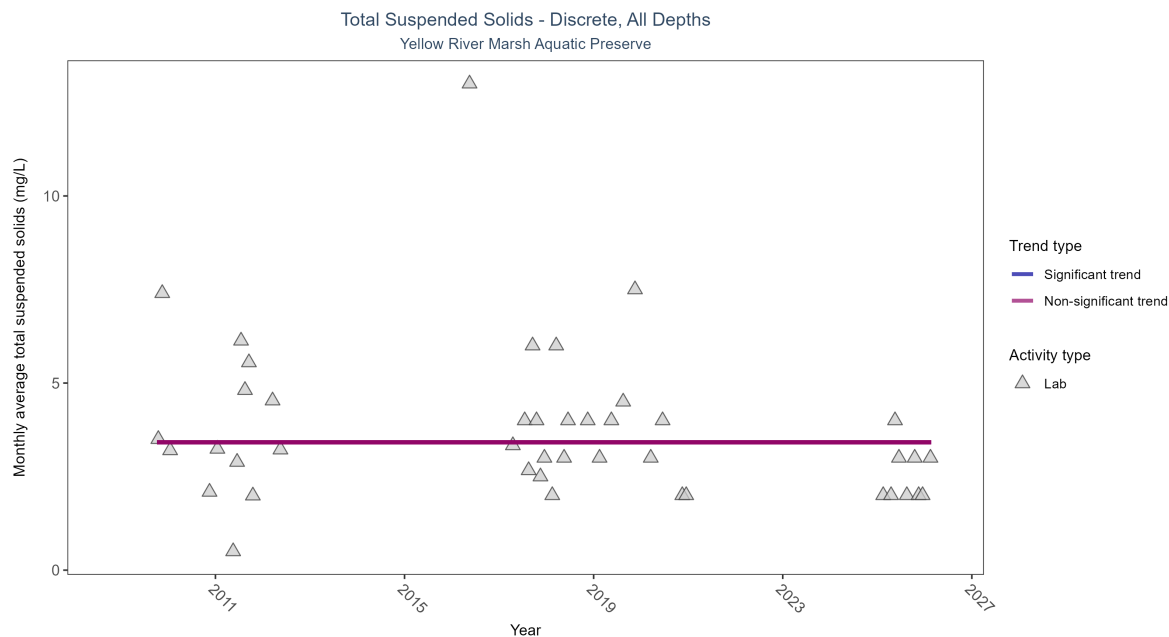


Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Total suspended solids showed no detectable trend between 2009 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	55	11	2009 - 2026	3.14	-0.07597	3.41786	0	0.7775

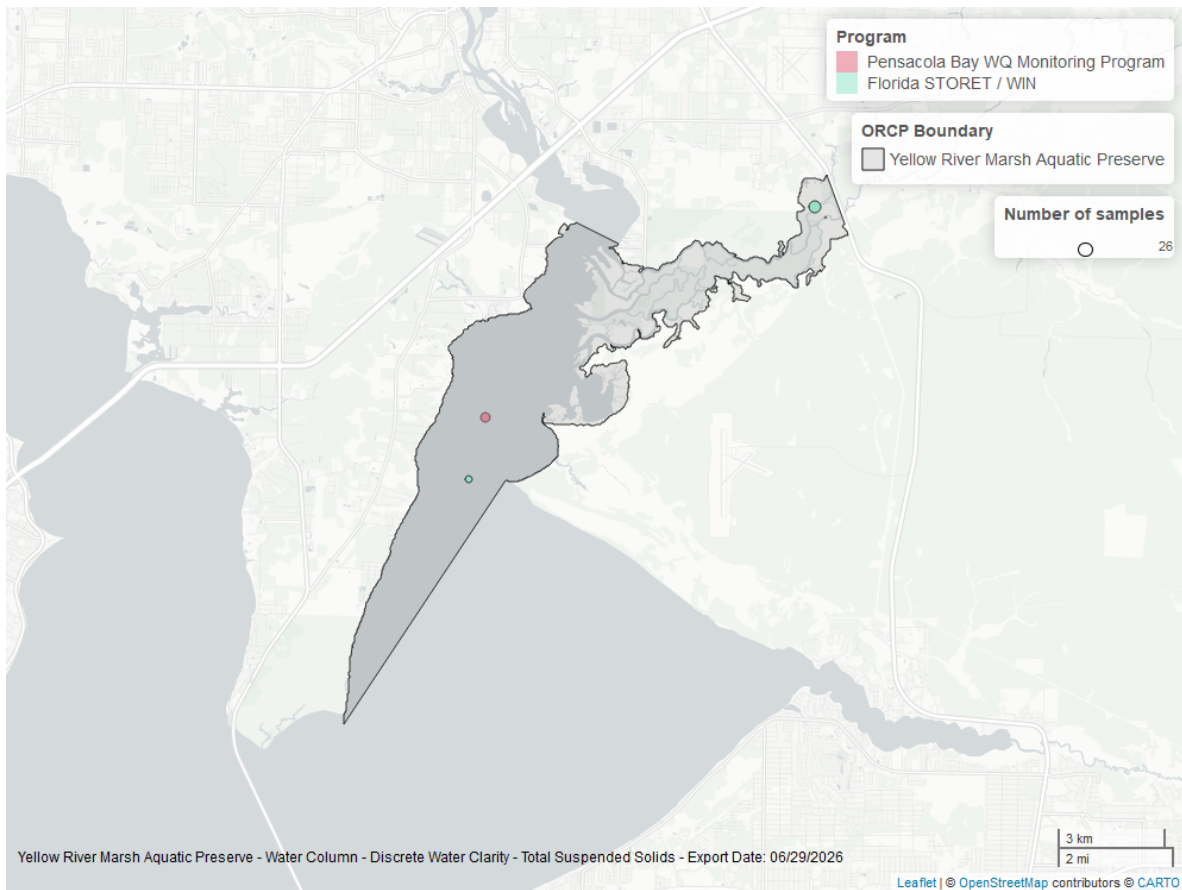


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

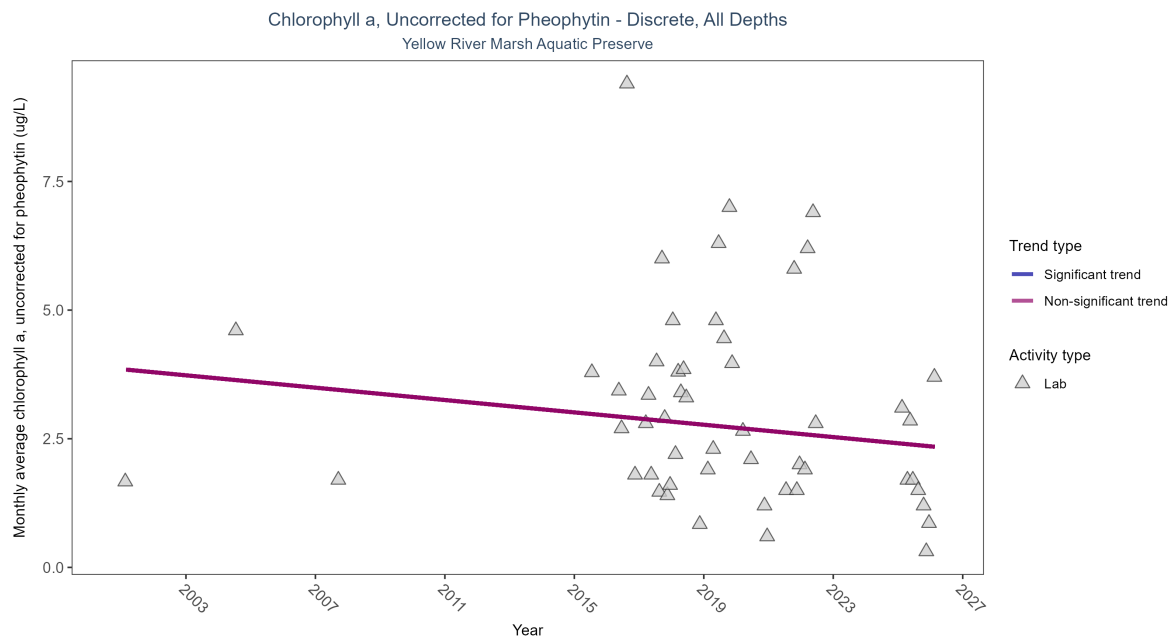


Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 2001 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	73	13	2001 - 2026	2.7	-0.05882	3.8525	-0.06	0.7104

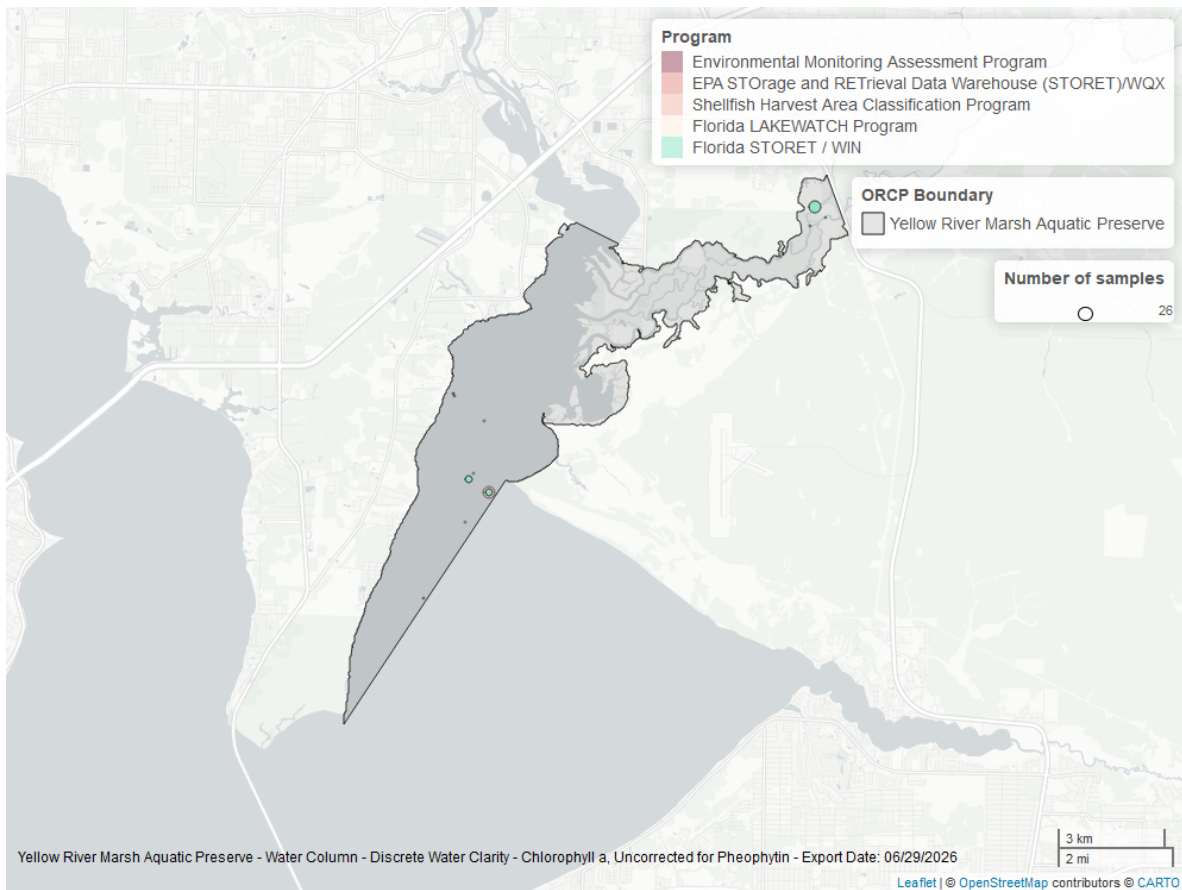


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

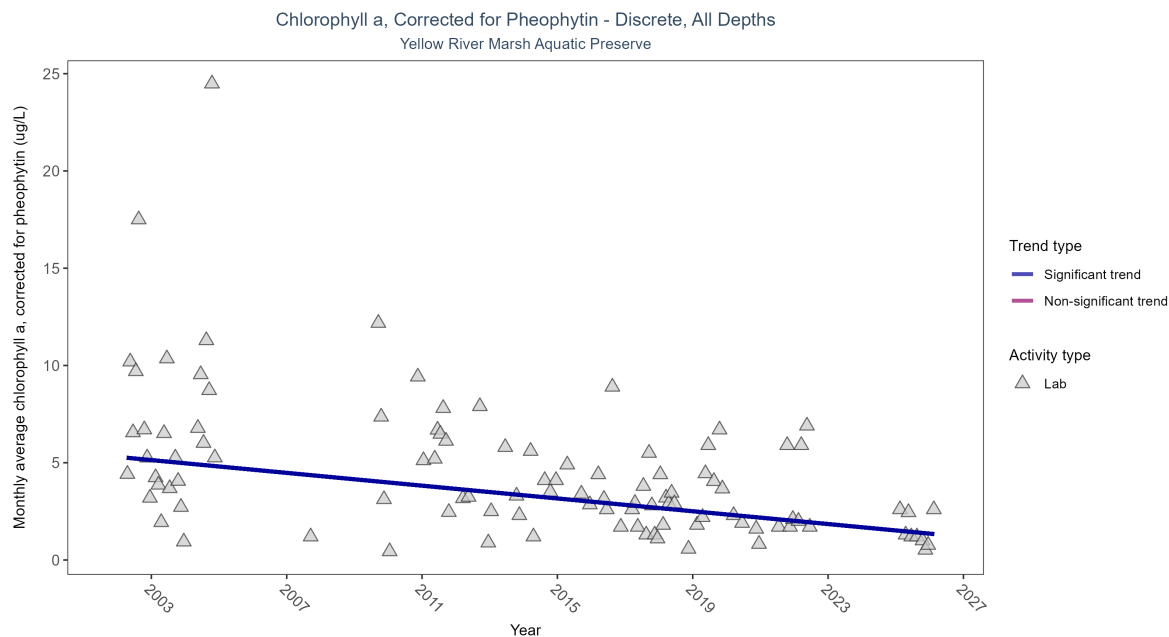


Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.16 ug/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	152	20	2002 - 2026	3.465	-0.35839	5.30057	-0.16438	0

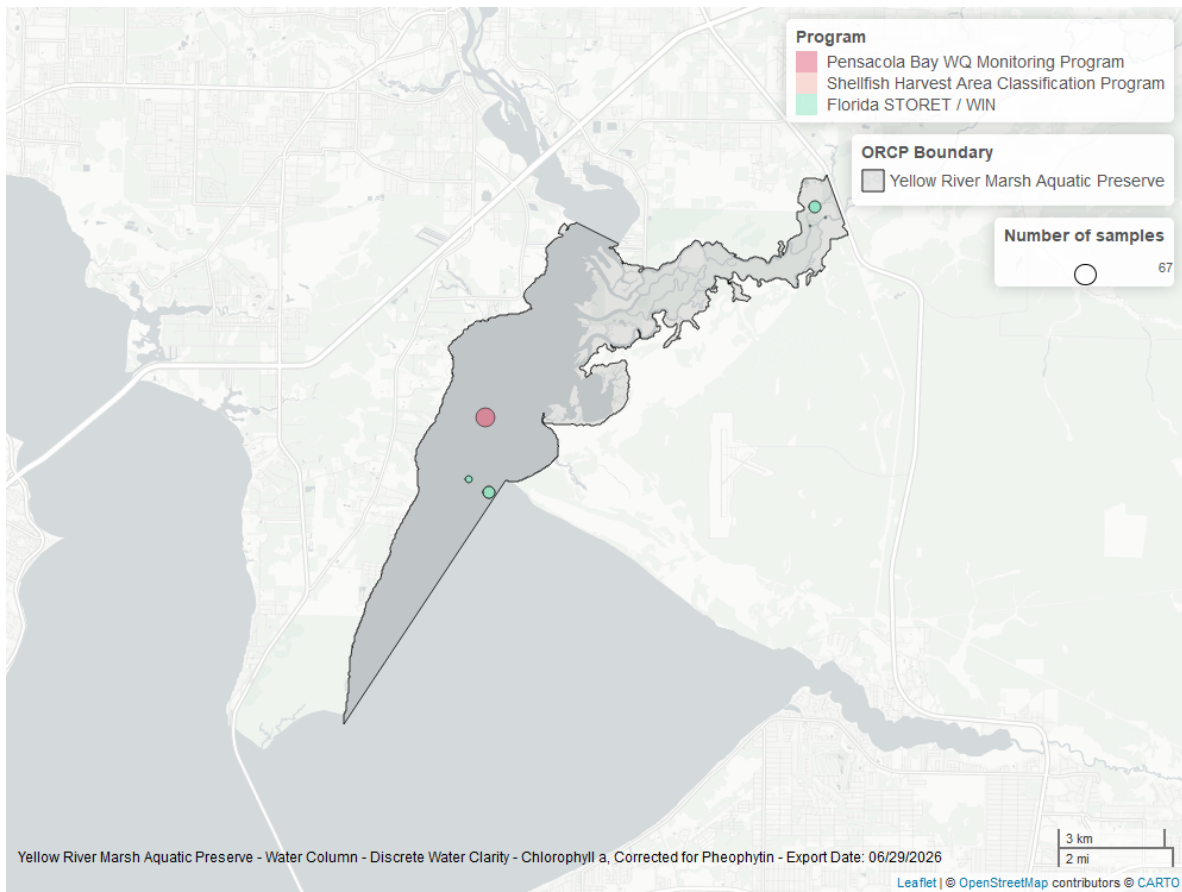


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

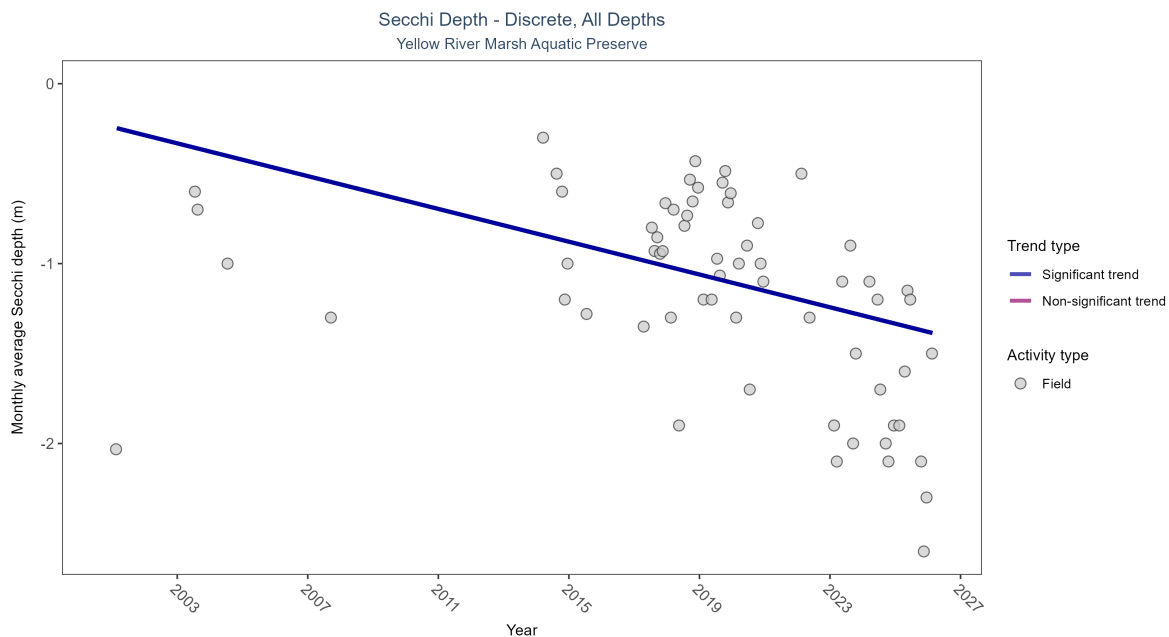


Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Monthly average Secchi depth became deeper by 0.05 m per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	247	15	2001 - 2026	-0.8	-0.35685	-0.24043	-0.04556	0.0018

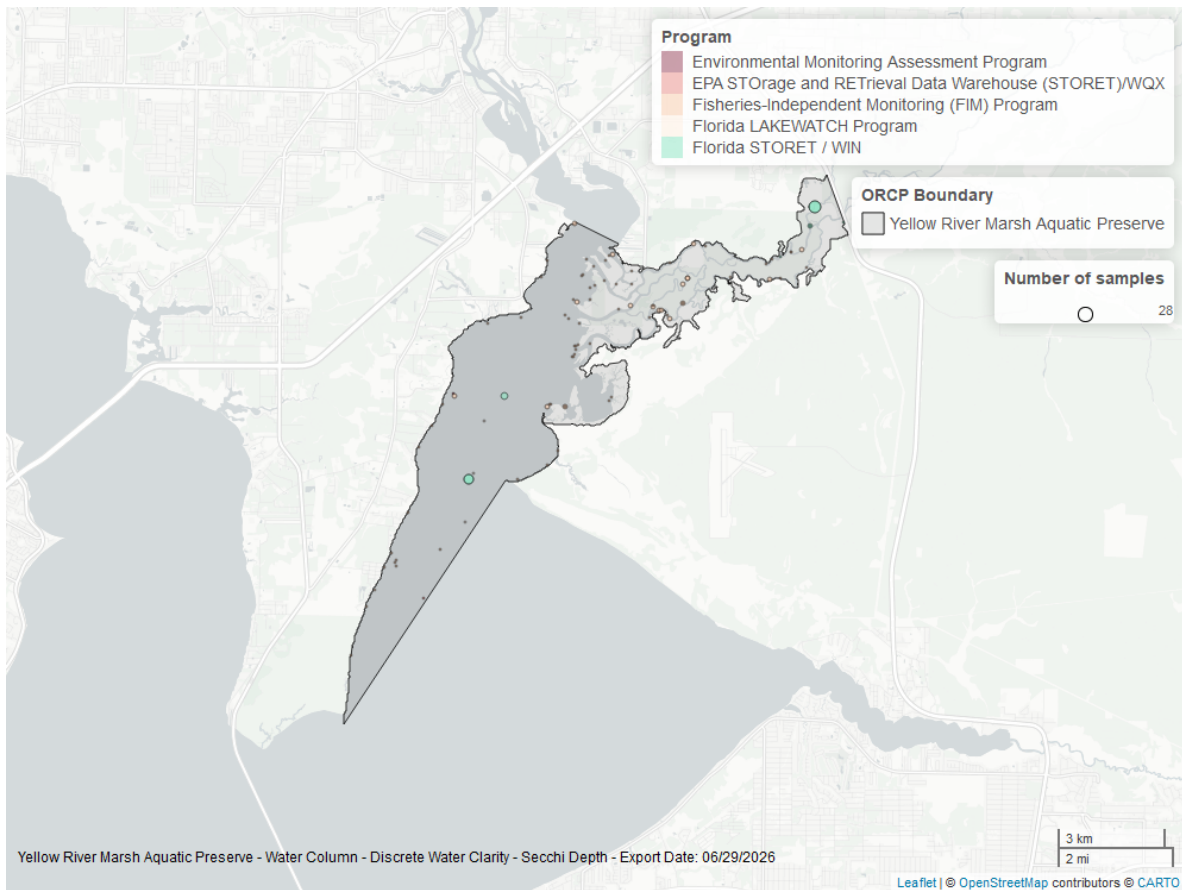


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

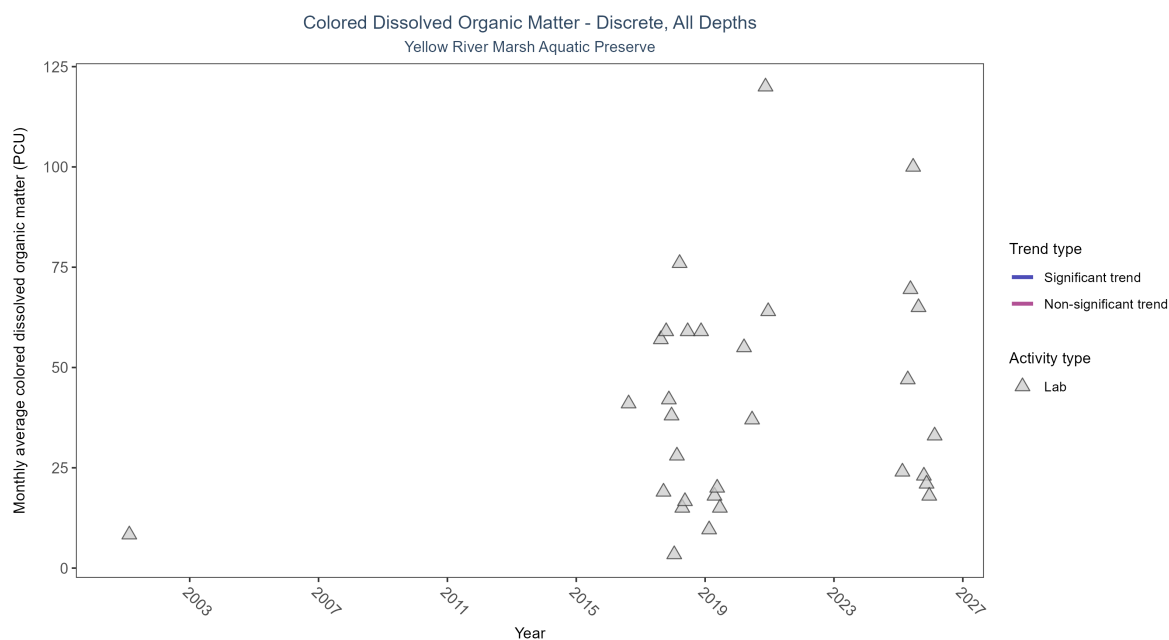


Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: There was insufficient data to fit a model for colored dissolved organic matter.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	37	8	2001 - 2026	28	-	-	-	-

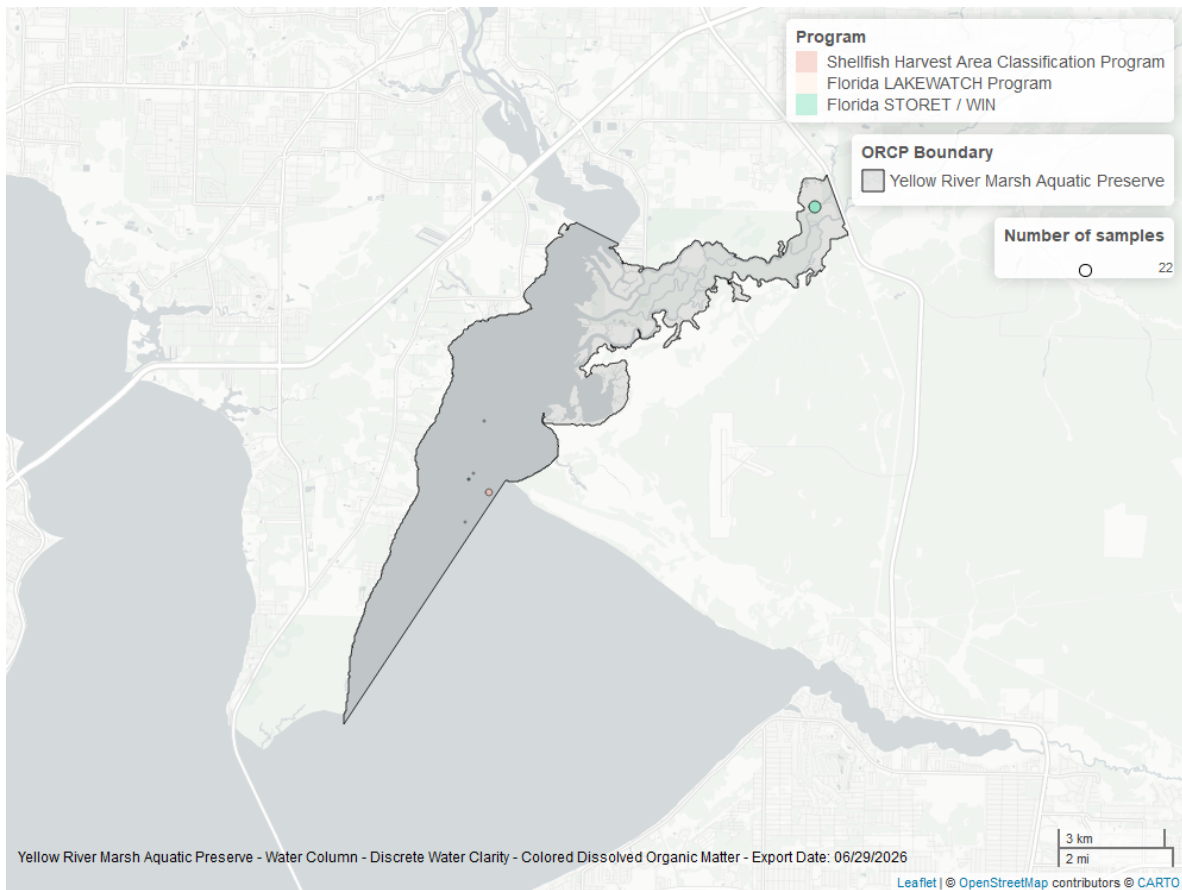


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.